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CloudSuite Industrial: v10 Managing Projects Training Workbook (10.18)

CloudSuite Industrial

February 8, 2019

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About this workbook

Welcome to this Infor Education course! We hope you will find this learning experience enjoyable and instructive. This Training Workbook is designed to support the following forms of learning:

- Classroom instructor-led training
- Virtual instructor-led training
- Self-directed learning

This Training Workbook is not intended for use as a product user guide.

Activity data

You will be asked to complete some practice exercises during this course. Step-by-step instructions are provided in this guide to assist you with completing the exercises. Where necessary, data columns are included for your reference.

Your instructor will provide more information on systems used in class, including server addresses, login IDs, and passwords.

Self-directed learning

If you are taking this course as self-directed learning, there may be instructor-recorded presentations and/or simulations available to assist you.

If instructor-recorded presentations are available, a hyperlink to the recording will be included on the first page of each corresponding Lesson.

If simulations are available, the demos and exercises throughout this Training Workbook will include hyperlinks that allow you to view and/or practice the execution of the demo or exercise in a simulated training environment.

Learning Libraries

Learning Libraries in Infor Campus include learning materials that are available to you online, anytime, anywhere. These materials can supplement instructor-led training, providing you with additional learning resources to support your day-to-day business tasks and activities.

Please note that if you accessed this Training Workbook directly via a Learning Library, you will not have access to the Infor Education Training Environment that is provided with all instructor-led and most self-directed learning course versions, as referenced above. Therefore, you will not be able to practice the exercises in the specific Training Environment for which the exercises in this Training Workbook were written.

Symbols used in this workbook



Hands-on exercise
("Exercise")



Your notes



Question



Instructor demonstration
("Demo")



Important note



Answer



Can be used for either
("Scenario" or "Discussion")



Critical note



Task simulation



For your reference



Course overview

Estimated time

.25 hours

Learning objectives

Upon completion of this course, you will be able to:

- Describe Projects.
- Describe the purpose of cost codes.
- Describe estimate projects, estimate project tasks, and estimate project resources.
- Explain the purpose of the copy projects activity.
- Identify characteristics of projects, project tasks, and project resources.
- List options to assist with planning for project resources.
- Define characteristics of project resource transactions.
- Describe project labor transactions.
- Describe the benefit of cross-referencing purchase orders to a project.
- Explain how to create purchase orders cross-referenced to a project.
- Describe aspects of jobs orders cross-referenced to a project.
- Describe the purpose of using vouchers/adjustments for projects.
- List details required when entering a voucher adjustment distribution record.
- Discuss how to ship product directly from a project.
- Identify ways to create advance ship notices for a project.
- List the three WIP Relief Method options.
- Identify product code requirements for WIP Distribution Accounts.
- Explain how to alter the WIP Relief Method for individual project tasks.
- State how to print/post project WIP relief.
- Describe the purpose and functionality of the Work Breakdown Structures form.
- Identify where in the application to set up revenue recognition for projects.
- Describe how to create revenue milestones for projects.
- State how to print/post project revenue milestones.
- Discuss how to set up invoicing for projects.
- Describe how to create invoice milestones for projects.
- Explain how to post project invoices.
- Describe how to generate payroll with labor transactions associated to a project.
- Describe Project Manager Home.
- Describe the information that can be accessed via the Navigator tab within Project Manager Home.
- Recognize import/export capabilities between Infor CloudSuite and Microsoft Project.
- Work with Projects.

Topics

- Course description and agenda

Course description and agenda

This course covers all phases of project planning and management including how to set up parameters, creating estimate projects, creating projects for which actual cost will accumulate, analyzing percent of completion to view costs, planning for and issuing project resources, applying labor, linking purchase orders or requisitions, cross-referencing to a job order, capturing costs, and performing cost analysis at various levels of detail. The course concludes with how to create revenue milestones and recognize revenue for a project.

This course is designed for version 10.18 of the software, but the vast majority of the lesson material applies to other versions as well.

Prerequisite courses

- CloudSuite Industrial: Navigating the User Interface
- CloudSuite Industrial: Foundation 101 (recommended, but not required)
- CloudSuite Industrial: Using Job Orders
- CloudSuite Industrial: Costing Foundation
- CloudSuite Industrial: Using Purchase Orders and Requisitions
- CloudSuite Industrial: Using Accounts Payable
- CloudSuite Industrial: Using Accounts Receivable

Course duration

8 hours

Prerequisite knowledge

To optimize your learning experience, Infor recommends that you have the following knowledge prior to attending this course:

- Knowledge of foundational concepts and how the data is structured and organized
- Ability to navigate the user interface

Audience

- Customer User
- Pre-Sales Consultant
- Business Consultant
- Technical Consultant
- Support
- System Administrator

System requirements

- CloudSuite Industrial® Training Environment

Reference materials

Reference materials are available from the following locations:

- CloudSuite Help menu

- Infor Xtreme®

Course agenda

Lesson	Lesson title	Learning objectives	Time Required
Course overview		Review course expectations.	.25 hours
1	Introduction to Projects	• Describe Projects. • Describe the purpose of cost codes.	.25 hours
2	Estimate projects	Describe estimate projects, estimate project tasks, and estimate project resources.	.5 hours
3	Projects	• Explain to the purpose of the copy projects activity. • Identify characteristics of projects, project tasks, and project resources.	.5 hours
4	Planning and projects	List options to assist with planning for project resources.	.25 hours
5	Project transactions	• Define characteristics of project resource transactions. • Describe project labor transactions.	.5 hours
6	Purchasing and projects	• Describe the benefit of cross-referencing purchase orders to a project. • Explain how to create purchase orders cross-referenced to a project.	.5 hours
7	Job orders and projects	Describe aspects of jobs orders cross-referenced to a project.	.5 hours
8	Vouchering and projects	• Describe the purpose of using vouchers/adjustments for projects. • List details required when entering a voucher adjustment distribution record.	.25 hours
9	Shipping and projects	• Discuss how to ship product directly from a project.	.25 hours

Lesson	Lesson title	Learning objectives	Time Required
		Identify ways to create advance ship notices for a project.	
10	Work in process (WIP) and projects	- List the three WIP Relief Method options. - Identify product code requirements for WIP Distribution Accounts. - Explain how to alter the WIP Relief Method for individual project tasks. - State how to print/post project WIP relief.	.5 hours
11	Work breakdown structures (WBS) and projects	Describe the purpose and functionality of the Work Breakdown Structures form.	.25 hours
12	Revenue recognition and projects	- Identify where in the application to set up revenue recognition for projects. - Describe how to create revenue milestones for projects. - State how to print/post project revenue milestones.	.75 hours
13	Invoicing and projects	- Discuss how to set up invoicing for projects. - Describe how to create invoice milestones for projects. - Explain how to post project invoices.	.75 hours
14	Payroll and projects	Describe how to generate payroll with labor transactions associated to a project.	.25 hours
15	Project Manager Home	- Describe Project Manager Home. - Describe the information that can be accessed via the Navigator tab within Project Manager Home.	.25 hours
16	Microsoft® Project	Recognize import/export capabilities between Infor CloudSuite and Microsoft Project.	.5 hours
17	Course review	Work with Projects.	1 hour
Course summary		Debrief course.	.25 hours

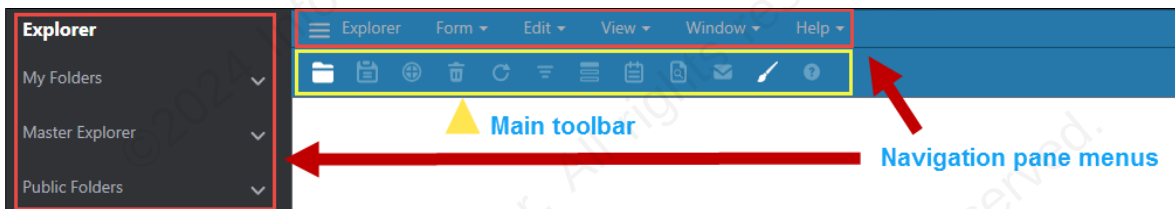
Appendices

This section contains information that is not part of the instructional content of this course, but provides additional related reference information.

Appendix	Appendix title	Content description
Appendix A	Learner user accounts	This appendix provides a reference for student login credentials.
Appendix B	Reports and utilities	Lists and describes project reports and utilities
Appendix C	Accounting references	Lists and describes accounting references to the Project Control Distribution Journal
Appendix D	Project change order tracking	Describes how to track changes for projects
Appendix E	Updates made to this version of the workbook	This appendix provides updates on content changes.

User interface

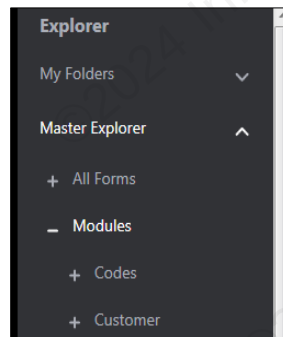
Standard user interface elements include the navigation pane menus and the main toolbar.



Navigation pane menus and main toolbar

Navigation pane menus

You can browse all available folders through the navigation pane menus and submenus. Click the down arrow for each of the navigation pane menu, this allows you to explore the submenus contained beneath.



Navigation pane menus and submenus

For information on how to pin submenus to the navigation pane, please refer to the Creating Form Personalizations workbook or ILT. For example, if you know that you will be using the Customer Service submenus frequently, you can display the menus at the top level, by pinning them to the navigation pane.

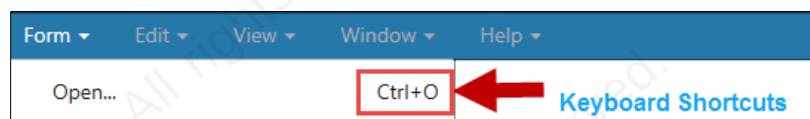
Main toolbar

The main toolbar contains icons that allow you to quickly complete commonly used tasks, such as saving data or closing a form. You can position your mouse pointer over an icon on the toolbar to reveal its description and functionality.

To view a list of all toolbar icons and for steps on how to add/remove icons from the default toolbar, See Appendix B. **Note:** Availability to make hiding buttons visible is not available for Web users.

Keyboard shortcuts

Some navigation pane submenus display available keyboard shortcuts that can be used to save multiple steps when completing certain tasks.



The following tables display the available shortcut keys and their actions.

Navigating forms

Action	Menu Path	Smart Client Shortcut	Web Client Shortcut	Icon	Button Default
Close current form	Form > Close and Cancel Changes	Ctrl + F4	N/A		Hidden
Open Select Form	Form > Open	Ctrl + O	Ctrl + O		Visible
Open workspaces	Form > Workspaces	Ctrl + W	N/A		
Switch view to next open form		Ctrl + Tab	N/A		

Navigating records

Action	Menu Path	Smart Client Shortcut	Web Client Shortcut	Icon	Button Default
Move to first editable field of form	View > Home Cursor	Ctrl + Home	Ctrl + Home		
Move to the next field		Tab	Tab		
Move to the previous field		Shift + Tab	Shift + Tab		
Move to the next record	Actions > Next	F8	F8		Hidden
Move to the previous record	Actions > Previous	F7	F7		Hidden
Switch the next collection on the form to become the current collection in a form that contains multiple collections.	View > Activate Next Collection	Ctrl + F8	N/A		Visible
Hide/Show 1 st Splitter Pane. Toggles back and forth between form and dual view.	View > Hide/Show 1 st Splitter Pane	Ctrl + 1	Ctrl + Shift + 1		
Hide/Show 2 nd Splitter Pane. Toggles back and forth between grid and dual view.	View > Hide/Show 2 nd Splitter Pane	Ctrl + 2	Ctrl + Shift + 2		

Editing records

Action	Menu Path	Smart Client Shortcut	Web Client Shortcut	Icon	Button Default
Activate the drop-down list with an implied asterisk (i.e. wildcard) after the entered text in the field to then populate the list with data entries that begin with the entered text.		F2	F2		
Add a new record Note: The Actions menu is only available when a form is open.	Actions > New	Ctrl + N	N/A		Visible
Add value for current field	Edit > Add	Ctrl + A	N/A		Hidden
Copy (only works to copy a record with web client)	Actions > Copy	Ctrl + C	Ctrl + C		
Cut		Ctrl + X	Ctrl + X		
Paste		Ctrl + V	N/A		
Undo	Edit > Undo	Ctrl + Z	N/A		
Delete record	Actions > Delete	Ctrl + D	Ctrl + D		Visible
Find value for current field (only available for certain drop-down fields)	Edit > Find Value for Current Field	Ctrl + F	N/A		Hidden
Display details for current field drill-down (only available for certain drop-down fields)	Edit > Details for Current Field	Ctrl + L	N/A		Hidden
Save changes (only available when a form is open)	Actions > Save	Ctrl + S	Ctrl + S		Visible

Filtering records

Action	Menu Path	Smart Client Shortcut	Web Client Shortcut	Icon	Button Default
In Filter Place mode, cancel the filter in place and return to the collection previously displayed	Actions > Filter > Cancel in Place	F3	F3		
In Refresh/Run mode, begin filter in place (i.e. clear the collection and go back to Filter In Place mode) In Filter In Place Mode, execute in place (i.e. run the filter)	Actions > Filter > Begin in Place Actions > Filter > Execute in Place	F4	F4		Visible
In Refresh/Run mode, refresh the current collection. In Filter In Place Mode, clear the filter in place.	Actions > Refresh Actions > Filter > Clear in Place	F5	F5		Visible
In Refresh/Run mode, refresh the current record.	Actions > Refresh current	Ctrl + F5	Ctrl + F5		Visible
In Refresh/Run mode, repeat Find Value in Collection	Edit > Repeat Find	Ctrl + F2	Ctrl + F2		
In Refresh/Run or New mode, open associated query form to specify filter criteria.	Actions > Filter > By Query	Ctrl + Q	N/A		
Open field level help topic	Help > Current Field	F1	N/A		
Toggle design mode on and off.	Edit > Design Mode	Ctrl + E	N/A		Visible
Print	Print	Ctrl + P	N/A		

Field types

The following tables display the field types available with a brief description:

Field type	Description
Required, system-generated, and read-only fields	
Order Date: 7/20/2013 15	- Any field that is required will have a red star - A system-generated field will have a green star - A field with no border around the data is read-only
Standard GUI fields	
☒ Control Point	Check box indicating Yes or No
☒ Female ☐ Male	Radio button to select one option from multiple choices
Order Type: Regular	Drop-down list to select one option from multiple choices

Icon name differences

Here are the names of the different icons in the smart client and web client.

Smart Client	Web Client
Open a form	Open
Save	Save
Create a new object in the current collection	New
Delete an object in the current collection	Delete
Refresh the current collection	Refresh
Filter in Place	Filter in Place
Get more rows in the current collection	Get more rows
Display notes for the current object	Notes
Display documents for the current object	Documents
Send email for the current object	Send email for the current object
Export the data in the current collection to excel	Export to excel
Enter or exit design mode on this form	Open designer in a new window
Translate	Translate

Smart Client	Web Client
Get help on the current form	Help Current Form



Lesson 1: Introduction to Projects

Estimated time

.25 hours

Learning objectives

In this lesson, you will:

- Describe Projects.
- Describe the purpose of cost codes.
- Describe Project Milestone Events.
- Describe Project Obligation Group Cost Codes.
- Describe Publication Subscribers

Topics

- Projects overview
- Projects interface
- Project Parameters – Default tab
- Project Cost Codes
- Project Codes – Project tab
- Project Milestone Events overview
- Project Obligation Group Cost Codes
- Publication Subscribers

Projects overview

Projects is a cost management tool. Projects allows you to capitalize costs for large, time-consuming projects or smaller, less time-consuming activities. The scope of a project could range from a government contract to internal tracking of development and design activities.

With Projects, your company can keep up to date and in a single location visibility for your plan as well actual for any project. Projects is a great cost collection tool for expenditures related to any project.

Here are some examples Projects can help with capturing costs for:

- Contracted projects
- Internal projects such as research and development, implementations, training, building improvements, software projects or facility moves

Without Projects, consolidated details would have to be kept track of manually. General ledger entries would have to be made and there is no consolidated way to get a comprehensive view of the project costs and status. With a little bit of setup in Projects such as cost codes, project types, tasks and product codes -you are off and running. Projects allows full visibility and tracking on your project, and reporting capabilities.

Projects is made up of project tasks. Each task could be a necessary step to complete the entire project. Furthermore, each project task can maintain a list of resources needed to perform the tasks.

The structure of Projects makes it possible to assign planned costs and commitments by project, task, and resource providing the ability to manage project activities and costs. This includes the capability to track and manage material, labor, overhead, and general and administrative (G & A) overhead that occurs at a higher level than a job order.

Actual costs accumulated to these areas can be captured on the project and project task. **Note:** For resources that are issued to the project and that are not linked to job orders, if the item is in the items table it must have a cost type of Actual for actual costs to be posted to the project. If the item's cost type is Standard, it is always issued to the project at its standard unit cost, even if the resource is linked to a PO line.

The actual cost accumulated on a project can be compared to the forecast cost of the project on the Percent of Completion forms. These forms can be viewed at both the project and project task level.

Once actual costs are accumulated on a project, they can be managed and controlled by supporting multiple user-defined cost codes and project accounts for material, labor, other, overhead, and G & A. Other cost/expense manipulations on a project occur during work in process (WIP) relief, revenue recognition, and invoicing directly from the project.

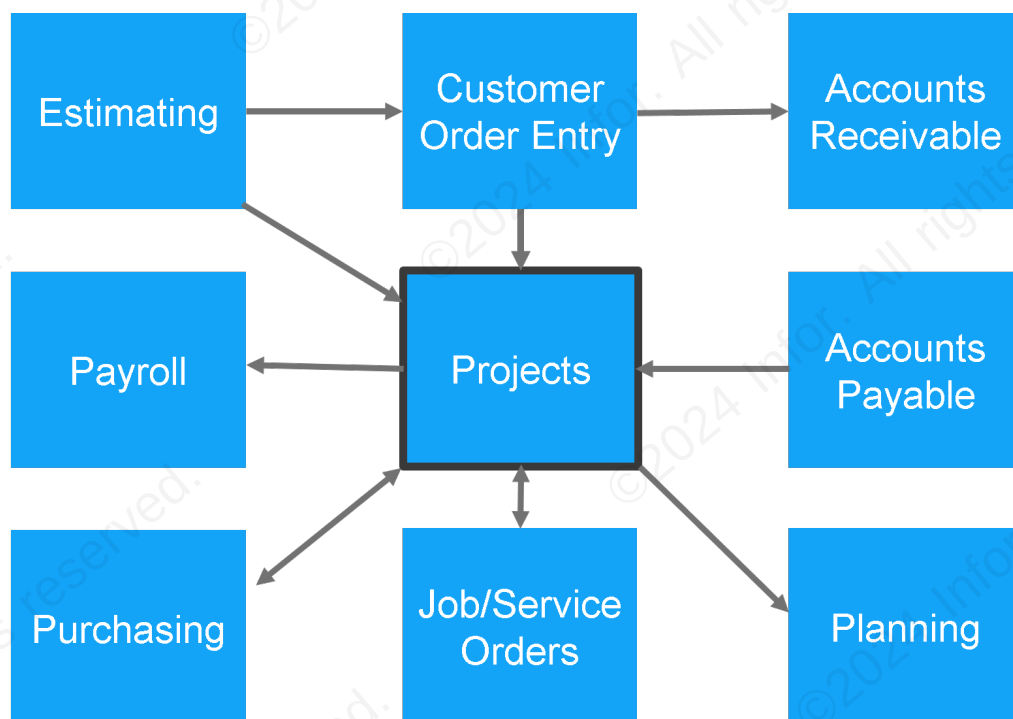


Projects interface

Project has many interfaces within the application. It interfaces with the following:

- Estimating
- Customer Order Entry
- Accounts Receivable
- Payroll
- Accounts Payable
- Purchasing
- Job/Service Orders
- Planning

The diagram below shows how Projects interfaces within the application.



Projects interface

- Estimate projects can be copied to Projects.
- Customer order lines can be cross-referenced to project tasks.
- Invoicing can take place in accounts receivable for customer order lines cross-referenced to project tasks.
- Direct labor hours posted to Projects can be used to generate payroll.
- Accounts payable expenses can be distributed to Projects.
- Purchase order requisitions and purchase orders can be cross-referenced from a project resource.
- Purchase order receipts can be captured to Projects.
- Job/Service orders can be cross-referenced from Projects.
- All costs for the job can be captured on Projects.
- Planning can capture project resources.

Project Parameters – Default tab

The Project Parameters form, **Default** tab defines how Projects should function. The **Project Prefix** field and expense accounts numbers can be accessed, updated, and entered on this form.

Information is entered in the **Estimate Prefix** field so that all new estimate projects with the same alphabetic and/or numeric prefix defaults for easy identification when reviewing reports.



Project cost codes

The **Project Cost Codes** form is used to input a user-defined code to categorize costs into labor, material, or other classes. These classes determine which project account is debited or credited when transactions are posted.

Individual cost codes determine whether overhead and/or G & A overhead should be calculated as a percentage of the project's cost. These percentages are defined on the Estimate Projects form.

A cost code with a Cost Class of Labor allows you to define the Default Cost Rate and a Default Budget Rate on this form. This dollar rate is a default used when adding cost codes and their forecasted/budgeted dollars to the Projects form and the Estimate Project Tasks form, % Complete tab.





Exercise 1.1: Log in to the application and enter project cost codes

In this exercise, you will log in to the application and you will create 5 new project cost codes for tracking on your project.

Before you begin:

- If you are using your own training environment, please contact your system administrator and have them give you a login to a fresh instance of the Demo database. The Demo database ships with the product, and all of the standard training courses for this application have been designed to use that database.

Exercise steps



Verify you are logged in to the training environment. If not, log in following instructions provided by your course instructor.

Note: For an SDL course, follow the instructions on the course **Lab On Demand** screen.

Part 1: Log in to the application

Web Client Login Instructions

1. Double-click **Train Web Client** on the desktop. The **Infor Business Cloud** window opens.
2. Type `<trainXX@gdeinfor2.com>` in the first field.
3. Type `G!oba!08` in the Password field.
4. Click **Sign In**.

5. The **Sign In** form opens.
6. Type <trainXX@gdeinfor2.com> in the **Username** field. See Appendix A for a list of usernames and passwords.
7. Type <trainXX> in the **Password** field.
8. Select <TrainXX> from the **Configuration** drop-down list.
9. Click **Sign In**. The application opens.

Smart Client Login Instructions

1. Double-click **Train Click Once Client** on the desktop. The **Sign In** form opens.
2. Type < trainXX@gdeinfor2.com > in the **Username** field. See Appendix A for a list of usernames and passwords.
3. Type trainXX in the **Password** field.
4. Select <TrainXX> from the **Configuration** drop-down list.
5. Click **OK**. The application opens.

Part 2: Enter project cost codes

1. Select **Master Explorer > Modules > Project > Files > Project Cost Codes**. The **Project Cost Codes** form opens.
2. Add a new record.
3. Type CM in the **Cost Code** field.
Note: To prepare any field in the system for text entry, first click in the text field.
4. Type *Construction Material* in the **Description** field.
5. Select **Material** from the **Cost Class** drop-down list.
Note: To display the options in any field, click the arrow in the field, and then select the options from the drop-down list.
6. Select the **Calculate Overhead** check box.
7. Select the **Calculate G & A** check box.
8. Save the record.
9. Add the cost codes, following steps 2-8, one at a time using the values in the table below for an external project.

Cost Code	Description	Cost Class	Calculate Overhead	Calculate G & A
CL	Construction Labor	Labor	Yes	Yes
GL	General Labor	Labor	Yes	Yes
MCE	Miscellaneous Expenses	Other	Yes	Yes

LM	Landscape Materials	Material	Yes	Yes
----	---------------------	----------	-----	-----

- 10. Save the record.
- 11. Close the **Project Cost Codes** form.

Product Codes – Project tab

The **Product Codes** form, Project tab defines asset accounts tied to a product code for journal entries affecting a project. These accounts post when project transactions are created during inventory transactions. For example, material issued for a job material is cross-referenced to a project.

The **Cost Code** field on the **Product Codes** form is the default cost code for items assigned with a specific Product Code.



Performance Obligation Events

Project Milestone Events overview

The new GAAP and IFRS guidance for 2018 states that we should recognize revenue on Projects in a way that is consistent with the transfer of goods or services to the customer.

Our user-defined Project Obligation Events can remind users to process Revenue and Invoice Milestones after time has elapsed, or labor hours or project costs have accumulated on a Project or Task.

Creating and Processing Project Milestone Events

There are three types of project milestone events that you can utilize: Project, Task Completion Event, and Performance Obligation Event. Create and manage notifications for these events, as well as fulfillment of performance obligations, by following these steps:

1. On the **Publication Subscribers** form, ensure that the appropriate users have subscribed to the Project Completion Alert, Project Task Completion Alert, and Project Obligation Completion Alert publications.
2. On the **Project Obligation Group Cost Codes** form, specify obligation groups and their associated cost codes. Once a group has been assigned to a performance obligation event, it cannot be deleted.
3. **Project Milestone Events** form. This form is used to manage milestone alerts for project completion events, project task completion events, and performance obligation events.
4. Specify this information in the Project group of fields:

Project

Select the project.

Enable MS for Project Completion

Select this field to enable notifications for all users subscribed to the Project Completion Alert publication, as well as any additional users specified in the User Notification field.

5. Specify this information in the Task Completion Event group of fields:

Enable MS for Task Completion

Select this field to enable notifications for all users subscribed to the Project Task Completion Alert publication, as well as any additional users specified in the User Name field.

Task

Ensure that the list of tasks is correct and complete. Corrections or additions can be made on the Project Tasks form.

Enable MS for Task Completion

Select this field to enable notifications when this specific task is complete. Notifications will be set to all users subscribed to the Project Task Completion Alert publication, as well as any additional users specified in the User Name field.

6. For each performance obligation, specify this information in the Performance Obligation Event group of fields:

Enable MS for Performance Obligation

Select this field to enable notifications for all users subscribed to the Project Obligation Completion Alert publication, as well as any additional users specified in the User Name field.

Invoicing

Select this field to indicate that an invoice milestone must be created and processed before the Processed field is selected, which indicates that obligation event is complete.

Obligation

Select the cost type of the obligation. Choose Labor, Material, Other, or All.

Obligation Group

Select one of the obligation groups that you created in step 2. Leave this field blank to apply the event to all obligation groups.

Project MS

Specify the type of milestone. Choose either Cost, Time Elapsed, Hours, or Percent Complete. If you select Time Elapsed, the system prompts you to create a background task. This background task will be used to trigger the event, rather than the Next MS Threshold amount that other milestone types use. If you fail to create a background task, the Project MS selection will change from Time Elapsed to Cost.

Recurring

Select this field if future event thresholds of this type will always be the same. When each recurring event is complete, a new event is automatically created with the same threshold. This field is disabled when Project MS is Time Elapsed.

MS Threshold

Meeting, or exceeding, the value specified here creates a milestone alert, which indicates that an invoice milestone can be created and processed. This field is disabled when Project MS is Time Elapsed.

Project Task

Select the project task with which to associate this obligation, or leave this field blank to avoid associating it with a specific project task.

7. After all sequences are created, click Save. The system calculates the Next MS Threshold amount for the initial sequence of each Project MS type, with the exception of Time Elapsed, which use a background task to control event triggering. However, for events marked as Recurring, the system creates a new obligation event with the same information and calculates the threshold for that event.
8. If you have added or removed project cost codes, click the Recalculate Obligations button to recalculate the accumulated totals and next scheduled amounts. You may also perform this task on the Project Obligation Group Cost Codes form. This step is only necessary if project or job transactions have occurred. We recommend that you complete all of your modifications before you click this button.
9. After an event is triggered, if the Invoicing field is selected, the system waits for an invoicing milestone to be processed before it marks the event as processed. If the Invoicing field is not selected, the system immediately marks the event as processed.
10. The system looks for the next event sequence with the same Obligation, Obligation Group, Project MS, and Project Task, then calculates its Next MS Threshold amount. If the event is recurring, the system creates a new sequence, then copies that recurring sequence's information into it.



Lesson 2: Estimate projects

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- Describe estimate projects, estimate project tasks, and estimate project resources.

Topics

- Estimate Projects
- Estimate Project Tasks
- Estimate Project Resources
- Estimate Project Tasks - % Complete tab

Estimate Projects

The **Estimate Projects** form allows you to enter an estimate project that can be manipulated to calculate the forecast cost for completing a project. An estimate project can be created by either manually adding the information on the **Estimate Projects** form or by cross-referencing an estimate order line item to an estimate project.

Estimate Projects – Header information and Address tab

The following table describes the fields on the **Estimate Projects** form, and **Address** tab:

Field	Description
Project	The project number defaults in this field to the next available estimate project number based on the Estimate Prefix in the Project Parameters form.
Date	The date the estimate project is created and defaults to the system is typed in this field.
Status	The status option selected in this drop-down list for an estimate project is always "Estimate." All estimating and cost calculations can be performed with a status of "Estimate."
Type	The type option selected in this drop-down list is a project classification. An estimated project must be designated as "Non-Billable" for an estimated project that does not have a customer.
Customer	A valid customer can be selected from the Customer field drop-down list.

Estimate Projects – General tab

The following table describes the fields on the **Estimate Projects** form, **General** tab:

Field	Description
Product Code	The option selected in the Product Code drop-down list indicates which project asset accounts to use for creating journal entries. This is a required field.
Estimate Project	The Estimate Project field shows the number referenced to a project This field is optional.
Project Mgr	The name of the project manager who controls the entire project is typed in this field.

Field	Description
Start Date	The date the estimate project should begin is typed in this field.
End Date	The completion date for the estimate project is typed in this field.
Contract Date	The date the contract is signed is typed in this field.
Contract	This field is used to enter the contract number assigned to a project.
Overhead Rate	The overhead rate to apply overhead costs for the entire estimate project is typed in this field; it is a required field.
G & A Rate	The G & A rate to apply a general and administrative overhead rate to be applied to the entire estimate project is typed in this field; it is a required field.





Exercise 2.1: Create an estimate project

In this exercise, you will create an estimate project entering values, using the following information.

Lake House Cycles approached your company about constructing a new bicycle racing track. Because each bicycle racing track has its own special features, the way to approach the cost of construction is to get extensive details about what Lake House Cycles expects for the bicycle racing track. The main contact for negotiations and information for the bicycle racing track is Jim Johnson.

At this point, the best thing to do is create an estimate project for the new bicycle racing track, referencing a Project Type of FIX-Fixed Contract. In normal situations, constructing a bicycle track takes about three months. Assume the start date is today.

Since your company will incur quite a few expenses that may not be obvious at this point, the estimate project calculations should include a 10% overhead rate and a 10% G & A rate.

You will collect more information from Jim Johnson later.

Exercise steps

1. Select **Master Explorer > Modules > Project > Estimate Projects**. The **Estimate Projects (Filter in Place)** form opens.
2. Run **Filter In Place**.
3. Add a new record.
4. Type *Bicycle Racing Track* in the field to the right of the **Project** field.
Note: This field is not labeled.
5. Select **FIX – Fixed Contract** from the **Type** drop-down list.
6. Select **12-Lake House Cycles** from the **Customer** drop-down list. The customer's address information displays.
7. Click the **General** tab.
8. Select **PROJ-FG-Finished Goods-Projects** from the **Product Code** drop-down list.
9. Type *Jim Johnson* in the **Project Mgr** field.
10. Type *<today's date>* in the **Start Date** field.
11. Type *<three months from today>* in the **End Date** field.
Note: The End User Type field defaults to RETAIL.
12. Type *10* in the **Overhead Rate** field.
13. Type *10* in the **G & A Rate** field.
14. Save the record. The estimate project number displays in the **Project** field.
15. Write down the estimate project number: _____.

Note: Do not close the **Estimate Projects** form; it is used in the next exercise.

Estimate Project Tasks

An estimate project is made up of estimate project tasks. The **Estimate Project Tasks** form defines the description of the different functions that need to be performed to complete the project.

An estimate project task can be created manually or by cross-referencing an estimate line to the estimate project task.

Estimate Project Tasks – Header

The following table describes the fields on the **Estimate Project Tasks** form header:

Field	Description
Project	The project number defaults in this field to the next available estimate project number based on the Estimate Prefix field in the Project Parameters form.
Date	Select or type the project creation date or accept the default value of the system date. This is a required field.
Status	The status selected in this drop-down list for an estimate project is always “Estimate.” All estimating and cost calculations can be performed with a status of “Estimate.”

Estimate Project Tasks – General tab

The following table describes the fields on the **Estimate Project Tasks** form, **General** tab:

Field	Description
Contact	This is the unique contact name you type for this task only.
Start Date	The date the task is to begin is typed in this field; it is a required field.
End Date	The date the task is to be completed is typed in this field; it is a required field.
Order	The customer order number displays if the project or estimate project is cross-referenced to a customer order or estimate order/line. The line number and release number display to the right of the Order field and these fields are display only.



Exercise 2.2: Create estimate project tasks

In this exercise, you will create an estimate project task.

After discussions with Jim Johnson, it was decided that there are five main tasks to complete the bicycle racing track. Lake House Cycles has already selected a prime location for the track. Each task should start after the previous task. As always, this is subject to change, weather permitting.

Exercise steps

1. Return to the **Estimate Projects** form.

Note: To return to any form, select the form name at the top of the screen. The **Estimate Projects** form opens. A list of estimate projects display in the grid view.

2. Select **<your estimate project>** from the grid view.
3. Click **Tasks**. The **Estimate Project Tasks (Linked)** form opens.
4. Add a new record.
5. Type *Survey the site* in the **Task Description** field.

Note: This field is not labeled and is located to the right of the **Task** field and the **Task** is **10**.

6. Type *<today's date>* in the **Start Date** field.
7. Type *<one week from today>* in the **End Date** field.
8. Save the record.

9. Add the estimate project tasks, following steps 4-8 to enter the estimate project tasks 20-50 one at a time using the values in the table. Before you start in the system, calculate and then write the start and end dates for each task. Assume the next task starts the same day the previous task ended to keep the example simple.

Note: Once you start to enter values, you may want to expand the left pane to see more of the grid and verify the dates you entered against this table.

Task	Task description	Contact	Your Start Date	Length of time to complete	Your End Date
20	Excavate the site		End date of task 10	Two weeks to complete	
30	Construct the track	Pete Trumble	End date of task 20	One and one half months	

40	Landscaping and adding finishing touches		End date of task 30	One week	
50	Test the track		End date of task 40	Duration of the project	

Note: Do not close the **Estimate Project Tasks (Linked)** form; it is used in the next exercise.

Estimate Project Resources

The **Estimate Project Resources** form allows you to identify any resources needed to perform an estimate project task. Resources can be referenced to Inventory, Purchase Orders, Purchase Order Requisitions, and Job Orders.

Estimate Project Resources – Task section

The following table describes the fields on the **Estimate Project Resources** form, **Task** section:

Field	Description
Item	The item number selected from the drop-down list can either be an inventoried or non-inventoried number. This is a required field.
Description	This field displays the description of the item, which defaults from the Item form if the item is stored in inventory.
Sequence	The system-generated sequence number displays. This number automatically increments in steps of one each time a new resource is added to the task. This field is helpful when reviewing an Estimate Project, Bill of Material.

Estimate Project Resources – General tab

The following table describes the fields on the **Estimate Project Resources** form, **General** tab:

Field	Description
Type	The Type field displays the type of item. The Type drop-down list options are: Material, Fixture, Tool, or Other. This is a required field.
Cost Code	The Cost Code field provides a drop-down list to associate the resource with similar cost resources. This allows for better cost analysis and visibility for the project. This is a required field.
Warehouse	The Warehouse field is used for item planning once the estimate project resource is copied to a project resource. This is a required field.
Due Date	The date displayed in this field is the date that planning uses for the resource. This is a required field.
Qty	The quantity of the resource item needed to perform the task is entered in this field. This is a required field.

Field	Description
U/M (unit of measure)	This field identifies the unit of measure (U/M) of the resource item. This is a required field.
Cost	This field displays read-only and identifies all of the costs associated with the item. This total cost can be updated on the Estimate Project Resources form, Cost tab.
Source	The Source drop-down list references the origin of the item. The reference options are: Inventory, Purchase Order, Job Order, Transfer Order, or Purchase Order Requisition.

Estimate Project Resources - Cost tab

The **Estimate Project Resources** form, **Cost** tab allows you to enter and view costs associated with each resource. The costs that can be maintained and updated include the following:

- Material Cost
- Labor Cost
- Fixed Overhead Cost
- Variable Overhead Cost
- Outside Cost

This tab also displays the actual Accum Values (Accumulated Values) of resources that were issued to the project.



Exercise 2.3: Add costs and allocate resources

In this exercise, you will add costs and allocate resources for the tasks you created in the previous exercises.

Exercise steps

Part 1: Add costs and allocate resources for Task 30

Note: Make sure all the resource costs are correct as you work through this exercise.

1. Return to the **Estimate Project Tasks (Linked)** form.
2. Select **Task 30-Construct the track** from the grid view.
3. Click **Resources** on the **Estimate Project Tasks (Linked)** form. The **Estimate Project Resources (Linked)** form opens.

Note: Task 30, Construct the track displays in the Task section of the form.

4. Select **GR-10000-Gravel, Grey** from the **Item** drop-down list. **Gravel, Grey** displays in the **Description** field.
5. Verify **Material** is in the **Type** field.

6. Select **CM-Construction Material** from the **Cost Code** drop-down list.
7. Type **13000** in the **Qty** field.
8. Verify **LB** is in the **U/M** field.
9. Verify **Inventory** is in the **Source** field.
10. Click the **Cost** tab.
11. Type **.30** in the **Material Cost** field.
12. Click the **General** tab.
13. Save the record.
14. Add a new record.
15. Add the estimate project resources, associated with Task 30, following steps 4-13 using the values in the table below.

Task 30							
Item	Description	Type	Cost Code	Qty	U/M	Source	Cost
TS-11000	Soil, Top	Material	CM	500,000	LB-Pound	Inventory	0.04
SA-10000	Sand	Material	CM	10,000	LB-Pound	Inventory	0.05
CB-10000	Block, Cinder	Material	CM	5,000	EA - Each	Inventory	0.20
RT-10000	Ties, Railroad	Material	CM	1,000	EA - Each	Inventory	1.00
RB-10000	Ramp, Bicycle	Material	CM	3	EA - Each	Inventory	360.00
WP-10000	Pond, Water Hazard	Material	CM	2	EA - Each	Inventory	210.00

16. Save the record.
17. Close the **Estimate Project Resources (Linked)** form. The **Estimate Project Tasks (Linked)** form displays.

Part 2: Add costs and allocate resources for Task 40

1. Select **Task 40-Landscaping** and adding finishing touches from the grid view.

2. Click **Resources**.
3. Type *Trees* in the **Item** field.
4. Press **Tab** to tab out of the field. A dialog box opens with the message: "Item that has [Item: Trees] does not exist. Do you want to Create/Edit Non-Inventory Item?"
5. Click **Cancel**.
6. Type *Trees* in the **Description** field.
7. Select **Material** from the **Type** drop-down list.
8. Select **LM-Landscape Materials** from the **Cost Code** drop-down list.
9. Type *20* in the **Qty** field.
10. Select **EA-Each** from the **U/M** drop-down list.
11. Select **Purchase Order** from the **Source** drop-down list.
12. Click the **Cost** tab.
13. Type *45.00* in the **Material Cost** field.
14. Click the **General** tab.
15. Save the record.
16. Add a new record.
17. Add the estimate project resources, associated with Task 40, following steps 3-16 using the values in the table below.

Task 40							
Item	Description	Type	Cost Code	Qty	U/M	Source	Cost
Shrubs	Shrubs	Material	LM	50	EA - Each	Purchase Order	10.00
Sod	Sod	Material	LM	5,000	SF-Square Foot	Purchase Order	.50

18. Save the record.
19. Close the **Estimate Project Resources (Linked)** form. The **Estimate Project Tasks (Linked)** form displays again.

Part 3: Allocate resources for Task 50

1. Select **Task 50-Test the track** from the grid view.
2. Click **Resources**.
3. Select **MB-10000-Bicycle, Model-180, Mountain, Black/White** from the **Item** drop-down list.

4. Verify **Material** is in the **Type** field.
5. Verify **MM-Manufactured Materials** is in the **Cost Code** drop-down list.
6. Type 5 in the **Qty** field.
7. Verify **EA-Each** is in the **U/M** field.
8. Select **Job** from the **Source** drop-down list.
9. Click the **Cost** tab.
10. Use the default **Cost** values for the item.
11. Click the **General** tab.
12. Save the record.
13. Close the **Estimate Project Resources (Linked)** form.

Estimate Project Tasks - % Complete tab

The **Estimate Project Tasks** form, **% Complete** tab shows the forecast and cost to complete amounts estimated for the task. The forecast cost can be added, updated, and/or deleted for individual cost codes. The information on this tab displays the amount of cost to complete, not the percentage of the project completed. You can view the project cost code and job cost code by selecting the corresponding radio button.

The bottom of the form displays the overhead and G & A amounts calculated based on the percentages set up on the **Estimate Projects** form. Cost codes come from the resources that were entered on the **Estimate Project Resources** form.

Estimate Project Tasks - grid columns

The following table describes the columns in the grid in the middle of the Estimate Project Tasks form:

Column	Description
Actual Cost	The Actual Cost column shows the total of all actual cost accumulated against a specific cost code for the estimate project. There should be no actual cost for an estimate project.
Forecast Cost (EAC) Estimate at Completion	The Forecast Cost (EAC) column displays the expected total costs of the resources by cost code. EAC is defined as the estimate at completion cost.
Cost to Complete	The cost needed to complete the task, displays in the Cost to Complete column. The cost to complete is forecast cost minus actual cost.
% Complete	The % Complete column indicates what percentage of cost has been completed for each cost code displayed. % Complete is calculated as the actual cost divided by the forecast cost. If the Forecast Cost field displays zero, then the % Complete column displays as 100%.
Budget	The budgeted amount for the project/project cost code is maintained in this column.





Exercise 2.4: Input estimated labor values

In this exercise, you will use the Estimate Project Tasks form - % Complete tab, and enter estimated labor costs.

Since labor cost will be a large expense incurred for constructing the bicycle racing track, this cost should be estimated utilizing the Estimate Project Tasks form - % Complete tab.

Currently, the cost information on the Cost Code line on the Estimate Project Tasks form - % Complete tab displays only the cost of the resource. The calculated OH and G & A costs display in the information below the table.

Exercise steps

Part 1: Enter budgeted labor costs for Tasks 10 and 20

1. Return to the **Estimate Project Tasks (Linked)** form.
2. Select the task **10-Survey** the site from the grid view.
3. Click the **% Complete** tab. A grid displays within the **Estimate Project Tasks (Linked)** form.
4. Click the **first row** in the grid in the **Cost Code** column.
5. Select **GL-General Labor** from the **Cost Code** drop-down list.
6. Scroll to the right to find the **Budget Unit Cost** field.
7. Click the **Budget Unit Cost** field.
8. Type **50** in the **Budget Unit Cost** field.
9. Click the **Budget Units** field.
10. Type **30** in the **Budget Units** field.
11. Click the **Budget** field. **Note:** The **Budget** field displays **1,500.00**.
12. Save the record.
13. Select the task **20-Excavate the site** from the grid view.
14. Add the estimated labor costs for Task 20, following steps 3-12 using the values in the table below.

Task	Cost Code	Budget Unit Cost	Budget Units	Budget should equal...
20-Excavate the site	GL-General Labor	40.00	100	4,000.00

15. Save the record.

Part 2: Enter budgeted labor costs for Tasks 30, 40 and 50

1. Select **Task 30-Construct the track** from the grid view.
2. Click the **first row** in the grid under the **Cost Code** field. This is so you can add a new record.

3. Add a new record.
4. Select **CL-Construction Labor** from the **Cost Code** drop-down list.
5. Scroll to the right and locate the **Budget Unit Cost** field.
6. Click the **Budget Unit Cost** field.
7. Type *50* in the **Budget Unit Cost** field.
8. Type *300* in the **Budget Units** field.
9. Tab out of the field.
10. The **Budget** field displays **15,000.00**.
11. Save the record.
12. Select the respective task from the grid, following steps 1-11 to enter and then save the estimated labor costs one at a time for Tasks 40 and 50. Use the values from this table:

Task	Cost Code	Budget Unit Cost	Budget Units	Budget should equal...
40-Landscaping and adding finishing touches	GL-General Labor	25.00	40	1,000.00
50-Testing the track	GL-General Labor	25.00	26	650.00

13. Save the record.
14. Close the **Estimate Project Tasks (Linked)** form.
15. Close the **Estimate Projects** form.

Check your understanding



What is the purpose of creating an estimate project?





Where are cost codes used on estimate projects?





Lesson 3: Projects

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- Explain the purpose of the copy projects activity.
- Identify characteristics of projects, project tasks, and project resources.

Topics

- Project Parameters
- Project Parameters – Evaluation tab
- Project Types
- Projects
- Project Tasks
- Project Resources
- Copy Projects

Project Parameters – Evaluation tab

The **Project Parameters** form, **Evaluation** tab displays **Schedule Variance** and **Cost Code Variance** fields.

Schedule Variance – Lower Deviation % and Upper Deviation %

The value in the **Schedule Variance** field is the allowable percentage deviation (budgeted costs/actual costs) that a project can have and still be considered in a state of control in terms of the schedule. Each record in the **Project/Job Accumulated Cost by Period** form is evaluated for this condition, and the overall project is evaluated on the **Projects** form as of the current state. On evaluation of the state of the project, project task, or project resource, the project is:

- Considered in danger of becoming out of control if the Lower Deviation % is exceeded for a point in time; then a yellow warning symbol displays on the Project/Job Accumulated Cost by Period form for the appropriate record.
- Considered out of control if the project exceeds the Upper % Deviation; then a red X displays on the Project/Job Accumulated Cost by Period form.

If neither warning situation is present, a green check mark displays.

The data in the **Priority Level** field is a numeric value you assign as a warning priority between the Cost Code Variance and the Schedule Variance. The Schedule Variance may be assigned a “1” and the Cost Code Variance a “2.” However, other numeric values can be used, with the lower value being the higher priority. The Priority Level comes into play when both the Cost Code Variance and the Schedule Variance have the same warning level. Only one warning level displays (on the Projects, Project Tasks, and Project Resources forms), and the value in the Priority Level determines which warning displays if both have the same warning level. You can enter a value for both **Priority Level** fields that are the same. In this case, the Schedule Variance is given the priority.

The **Priority Level** rules are as follows:

If...	Then...
Both the Cost Code Variance and Schedule Variance fields display a “red” condition	The indicator displays the one with the highest priority level (the lowest number in the Priority Level field).
Both the Cost Code Variance and Schedule Variance fields display a “yellow” condition	The indicator also displays the one with the highest priority level (the lowest number in the Priority Level field).
Either the Cost Variance or the Schedule Variance field (one or the other) display a “red” condition	The field with the “red” indicator displays.
One field displays a “yellow” condition (and the other field has a “green” condition)	The field with the “yellow” variance displays.

If...	Then...
Both the Cost Code Variance field and the Schedule Variance field have a “green” condition	A “green” check mark displays with the text “In Control.”

Cost Code Variance – Lower Deviation % and Upper Deviation %

Cost Code Variance is the allowable percentage deviation (budgeted/actual costs in terms of the cost codes) that a project or group of projects is allowed to deviate from within a period. Each record in the selection set for a cost code or group of cost codes is evaluated for this condition and the overall group is evaluated as follows:

If...	Then...
The value in the Cost Code Variance field exceeds the value in the Lower Deviation % field in the group	A yellow warning symbol displays on the Project/Job Period Costs form for the appropriate record.
The value in the Cost Code Variance field is exceeded for the current period	A yellow warning symbol displays for a project, project task, or project resource evaluated against all cost codes as a group.
The value in the Upper Deviation % field is exceeded	A red X displays on the Project/Job Period Costs form for the appropriate record.
Neither warning situation is present	A green check mark displays.
The value in the Cost Code Variance field exceeds the value in the Lower Deviation % in the group	A yellow warning symbol displays on the Project/Job Period Costs form for the appropriate record.

The **Priority Level** field contains a numeric value you assign as a warning priority between the **Cost Code Variance** and the **Schedule Variance** fields. The schedule variance may be assigned a “1” and the cost code variance a “2.” However, other numeric values can be used, with the lower value being the higher priority. The Priority Level field comes into play when both the **Cost Code Variance** and the **Schedule Variance** fields have the same warning level. Only one warning level displays (on the **Projects**, **Project Tasks**, and **Project Resources** forms), and the Priority Level field value determines which warning displays if both have the same warning level. You can enter a value for both **Priority Level** fields that are the same. In this case, the Cost Code Variance is given the priority.

Project Types

The **Project Types** form is used to input user-defined codes that represent different project types or classifications.

Pricing categories

Project types are required. They are classified by the following pricing categories:

Pricing category	Description
Fixed Price	A fixed-priced project should have a specific contract revenue amount agreed upon. Invoices should be based on planned revenue.
Calculated	Invoices should be based on the actual costs accumulated on a project that are in this category.
Non-Billable	There should be no revenue associated with a project that falls under the non-billable pricing category.



Projects

The **Projects** form allows you to enter a project that can be set up to calculate the forecast cost for completing the project. The project can be cross-referenced from the **Customer Order Lines** form or copied from an estimate project. Projects can also be manually created.

Projects – Header information

The following table describes the fields and check boxes on the **Projects** form header

Field/check box	Description
Project	This field displays the project number that defaults to the next available project number based on the Project Prefix in the Project Parameters form.
Customer	This field displays a valid customer associated with the project. A customer number is not required for internal projects, but a customer number is required for project invoice milestone posting. If a customer number is not used, the project type must be non-billable.

Projects – General tab

The **Projects** form, **General** tab defines criteria relevant to the project. The following table describes the fields on the **Projects** form, **General** tab:

Field	Description
Status	This field displays the project status which is selected from one of the following drop-down list options: Active, Inactive, or Complete. This is a required field. • All project cost accumulation can be performed with the status of “Active.” • Cost accumulation cannot be performed against a project with a status of “Inactive.” • A status of “Complete” specifies that a project is no longer active.
Type	The Type field option identifies the project classification. This is a required field. Note: A project with no customer must be identified as Non-Billable.
Product Code	The Product Code field option identifies which project asset accounts are used for creating journal entries. This is a required field.
Date	The Date field identifies the date the project is created and defaults to the system date. It is a required field.

Field	Description
Late Alert	The system selects the Late Alert check box to indicate when a project that has one or more resources is either projected to be late or are already late. If the project is on schedule, the system clears the check box. A resource is considered late under the following conditions: - its projected date is after the due date and the quantity issued for the resource is less than the resource quantity. - when the projected date is blank, the resource's current date is on or after the due date and the quantity issued for the resource is less than the resource quantity. If you are using MRP, the system compares the resource due date to the current date. This check box only indicates whether the resource is already past due. It does not indicate whether the resource is projected to be on schedule.
Contract Date	This is the date the contract was signed.
Contract	This is the contract number assigned to the project.
Estimate Project	This is the estimate project number referenced to a project.
Overhead Rate	The overhead rate percentage is typed in this field to apply overhead costs for the entire project.
G & A Rate	The G & A percentage rate is typed in this field to apply the general and administrative overhead rate to the project. This is a compounded overhead rate.
End User Type	This field can be used in determining a discount in conjunction with the product code of the line/release. Discounting can be made by customer type or by product code.
Track Changes	The Track Changes check box is only enabled when the Project Types form has "Tracking" selected. If selected, a warning message displays with the message, "Once Tracking is turned on it cannot be turned off." The option is given to "Cancel" or "Continue." Changing the Project Type field does not turn off Track Changes even if the new type is not set for Tracking.
Change	The Change field displays the current change number for the project; this is a read-only field.
Change Status	This field is not labeled. This read-only field shows the number of the most recent change order to a project.
Detail	The Detail field displays the tasks and resources in a tree view.

Field	Description
Project Mgr	The name of the project manager who controls the entire project is typed in this field.



Change Notes button - The Change Notes button is disabled when Track Changes check box is not selected. This allows the addition of notes to the Project Change Orders form



Appendix D in Training Workbook

See Appendix D: Project Change Order Tracking to learn more about tracking changes on a project.

Projects – % Complete tab

The **Projects** form, **% Complete** tab displays the percentage of the forecast cost incurred on a project. These costs are broken down by cost code. The information on this tab is display only and no changes can be made to the values in the **Actual Cost**, **Forecast Cost**, or **Cost to Complete** fields. Adjustments can only be made on the **Project Tasks** form, **% Complete** tab.

The other values that display are the **Overhead Rate** field and **G & A Rate** field.

Projects - % Complete – grid columns

The following table describes the columns in the grid on the **Projects** form, **% Complete** tab:

Column	Description
Actual Cost	The Actual Cost column displays the total of all actual costs by cost code.
Forecast Cost	The Forecast Cost column displays the expected total cost to complete the project by cost code.
Budget Cost	The Budget Cost column displays the original total cost to complete the project by cost code.
Cost to Complete	The Cost to Complete column displays the total of all costs needed to complete project tasks by cost code. The Cost to Complete field displays the value of Forecast Cost column minus Actual Cost column.
% Complete	The % Complete column indicates what percentage of cost has been completed for each cost code displayed. The % Complete field value is calculated as the actual cost divided by the forecast cost.



Useful project reports

Useful reports are Project Margin Analysis Report and Project Transactions Report. Both reports are accessed by selecting **Master Explorer > Modules > Project > Reports**.

Projects – % Complete Details tab

The **Projects** form, **% Complete Details** tab differs from the information found on the **% Complete** tab. The information that displays is broken down by cost code and monthly period, providing more detail than is on the **% Complete** tab.



Project Tasks

A project is made up of project tasks. The **Project Tasks** form defines the description of the different functions that need to be performed to complete the project.

Project Tasks – Header

The following table describes the fields on the **Project Tasks** form, header information:

Field	Description
Task	The Task field displays a number that is associated with a specific function of the project. The number is automatically incremented in steps of 10. This number can be overwritten. A description of the task is entered in the blank text box to the right of the number.

Project Tasks – General tab

The following table describes the fields and check boxes on the **Project Tasks** form, **General** tab:

Field	Description
Status	This field displays the status for a project task. The options are: Active, Inactive, or Complete. • All project cost accumulation can be performed when the status is “Active.” • Cost accumulation cannot be performed against a project task with the status of “Inactive.” • A status of “Complete” specifies that a project task is no longer active.
Task Date	The Task Date field shows the date the task is entered into the system.
Start Date	This field displays the date the task is to begin. This is a required field.
End Date	This field displays the date the task is to be completed. This is a required field.
Projected (date)	The field displays the date when the planning or scheduling functions estimate a job, customer order, or other demand can be completed, based on other demands and your current capacity. A date can be generated for various types of demands by the Scheduler or the APS system.
Late Alert	The system selects the Late Alert check box to indicate when a project that has one or more resources is either projected to be late or are already late. If the project is on schedule, the system clears the check box.

Field	Description
	A resource is considered late under the following conditions: - its projected date is after the due date and the quantity issued for the resource is less than the resource quantity. - when the projected date is blank, the resource's current date is on or after the due date and the quantity issued for the resource is less than the resource quantity. If you are using MRP, the system compares the resource due date to the current date. This check box only indicates whether the resource is already past due. It does not indicate whether the resource is projected to be on schedule.
Contact	This field displays the unique contact name for this task only.
Order	The order is referenced in the Order field if there is a cross-reference link between a customer order/line and a project task.



Useful project reports

Useful reports are Project Variances Report and Project Transactions Report. Both reports are accessed by selecting **Master Explorer > Modules > Project > Reports**.



Project Tasks - % Complete tab

The **Project Tasks** form, **% Complete** tab shows the forecast and cost to complete amounts planned for the task as well as the actual cost accumulated. The **Forecast Cost** and **Cost to Complete** columns can be added, updated, and/or deleted for individual cost codes at the task level. The information on this tab displays the amount of cost to complete, not the percentage of the project completed.

The lower portion of the form displays overhead and G & A overhead amounts calculated based on the percentages set up on the Projects form.

Project Tasks – grid columns

The following table describes the columns in the grid in the middle of the form:

Column	Description
Actual Cost	The Actual Cost column displays the total of all actual cost accumulated against a specific cost code for the project.
Forecast Cost (EAC)	The Forecast Cost column displays the expected total costs of the resources displayed by cost code.
Budget	The amount budgeted for the project/project cost code is maintained in this column.
Cost to Complete	The Cost to Complete column displays the cost needed to complete the task. The cost is determined by subtracting the value in the Actual Cost column from the value in the Forecast Cost column.
% Complete	The % Complete column indicates the percentage of cost completed for each cost code. The % Complete value is calculated by dividing the value in the Actual Cost column by the value in the Forecast Cost column. If the value in the Forecast Cost column is zero, then the % Complete column displays as 100%.



Project Resources

The **Project Resources** form allows you to identify any resources needed to perform a project task. Resources can be referenced to Inventory, Purchase Orders, Purchase Order Requisitions, and Job Orders.

Project Resources – Header

The following table describes the fields on the Project Resources form:

Field	Description
Item	The item selected from the Item drop-down list can be an inventoried or non-inventoried item. This is a required field
Description	This field displays the description of the item. This description defaults from the Item form if the item is stored in inventory.
Sequence	This field displays the sequence number which automatically increments in steps of one each time a new resource is added to the task. This can be helpful when reviewing a project job or bill of material.

Project Resources – General tab

The following table describes the fields on the **Project Resources** form, **General** tab:

Field	Description
Type	The options available in the Type field drop-down list are: Material, Fixture, Tool, or Other. This is a required field.
Cost Code	The Cost Code field option groups the cost of resources with the same cost code for better cost analysis and visibility for the project. This is a required field.
Qty	This field displays the quantity of the resource item needed to perform the task. This is a required field.
U/M (Unit of Measure)	This options in the U/M drop-down list identify the unit of measure. This is a required field.
Cost	This field identifies all the costs associated with the item. This total cost can be updated on the Project Resources form, Cost tab.
Warehouse	This field is used for project planning.
Due Date	This field displays the due date used for planning resources. This is initially derived from the project task start date.

Project Resources – Source tab

The **Source** field drop-down list references the origin of the item. The options are: Inventory, Purchase Order, Job Order, Purchase Order Requisition, SRO, or Transfer Order.



Multi-site users can source a transfer order from the main site to the supply site for project resources and transfer from warehouse to warehouse within a site. This will signal another site or warehouse to send the needed supplies.

Project Resources - Cost tab

The **Project Resources, Cost** tab allows you to enter and view costs associated with each resource. The costs that can be maintained and updated include the following:

- Material Cost
- Labor Cost
- Fixed Overhead Cost
- Variable Overhead Cost

- Outside Cost

This tab also displays the actual Accum Values (Accumulated Values) of resources that were issued to the project.

Copy Projects

The **Copy Projects** activity allows you to copy a project, an estimate project, or history project to a new or existing one.



The **Copy Projects** activity copies the estimate project, estimate project tasks, estimate project resources, revenue milestones, and invoice milestones.



Exercise 3.1: Create a new project by copying the estimate project

In this exercise, you will create a new project by copying the estimate project to a new project. Once the project is in place, actual cost accumulation can begin.

Great news! Lake House Cycles has accepted the estimate project and so you can transfer into a project.

Exercise steps

1. Select **Master Explorer > Modules > Project > Activities > Copy Projects**. The **Copy Projects** form opens.
2. Select **Estimate** from the **From** drop-down list.
3. Select **<your estimate project>-Bicycle Racing Track** from the **Estimate** drop-down list.
4. Verify **10-Survey** the site is in the **Starting Task** field.
5. Verify **50-Test the track** is selected in the **Ending Task** field.
6. Verify **Project** is in the **To** drop-down list.
7. Verify these check boxes are selected:
 - Copy Tasks
 - Copy Resources
 - Copy Revenue Milestones
 - Copy Invoice Milestones
8. Verify the **Preview** radio button is selected.
9. Click **Process**. A dialog box opens with the message, "5 Project Task(s) were displayed."
10. Click **OK**.
11. Select the **Commit** radio button.
12. Click **Process**. A dialog box opens with the message, "[Copy Projects] will be performed."
13. Click **OK**. A dialog box opens with the message, "5 Project Task(s) were processed."
14. Click **OK** again.
15. Write down the value in the **New Project** column: _____.
16. Close the **Copy Projects** form.



Exercise 3.2: Verify project information

In this exercise, you will verify project information.

Exercise steps

Verify that all information on the project is correct. The information should be identical to the estimate project that was copied to this new project.

Part 1: Review the Projects form

1. Select **Master Explorer > Modules > Project > Projects**. The **Projects (Filter In Place)** form opens.
2. Type **<your project number>** in the **Project** field.
3. Run **Filter In Place**. Now you are viewing only **your project** information.

Part 2: Review the Project Tasks form

1. Click **Tasks**. The **Project Tasks (Linked)** form opens.
2. Confirm that **five Project Tasks** display in the grid.
3. Close the **Project Tasks (Linked)** form.

Part 3: Review the Project Resources form

1. Click **Resources**. The **Project Resources (Linked)** form opens.
2. Confirm the **11 Project Resources** display.
3. Click the **% Complete** tab.

Note: The costs were copied over from the estimate project and display on the bottom of the form in the separated columns.

4. Close the **Project Resources (Linked)** form.



Exercise 3.3: Enter the contract number and revenue

In this exercise, you will enter the contract number on the project you created and indicate the WIP start date and revenue amount.

Now that Lake House Cycles has accepted our quote and signed a contract, the contract number needs to be included on the project and you need to enter the contract revenue amount.

The contract number is LHC – 1020.

Exercise steps

1. Return to the **Projects** form.
2. Click the **General** tab.

3. Select **<today's date>** from the **Contract Date** calendar.

4. Type *LHC-1020* in the **Contract** field.

Note: The Estimate Project number displays in the **Estimate Project** field.

5. Click the **Revenue** tab.

6. Type *100000* in the **Project Totals Planned** section **Total Revenue** field.

7. Save the record.

Leave the **Projects** form open as you will use it in Lesson 5.

Check your understanding



What is the difference between the % Complete tab and the % Complete Details tab on the Projects form?





What are the different ways that a project can be created?





Lesson 4: Planning and projects

Estimated time

.25 hours

Learning objectives

After completing this lesson, you will be able to:

- List options to assist with planning for project resources.

Topics

- Planning - Project Resources
- Material Planner Workbench
- Planning Detail

Planning – Project Resources

Resources for a project can be planned via Planning for any item that exists in the Items table. The quantity of a resource item is an independent demand used in planning.

The start date of the task is the date used as the requirement for the item. This date is identified in the **Start Date** field found on the **General** tab of the **Project Tasks** form.



Material Planner Workbench

The Material Planner Workbench offers an alternative method for generating supply orders that satisfy requirements without needing to run Material Requirement Planning (MRP) or the Advanced Planning and Scheduling (APS). The application considers the demands and supplies at all warehouses across the entire plan and creates planned supply orders to cover all instances where the on-hand inventory is below safety stock. The application places the item replenishment data on the Material Planner Workbench, where it can be reviewed, changes made, and firm it into purchase orders, purchase requisitions, jobs, production schedules, Master Production Schedule (MPS) orders, and transfer orders.



Planning Detail

The **Planning Detail** form allows you to view individual transactions created after you run the APS, MRP, or the MPS Processor. This form indicates the day you need each requirement. This form also allows you to verify planned orders prior to confirming them.



Planning Details is most useful in the grid view.





Exercise 4.1: Plan for all project resource items

In this exercise, you will activate, and run MRP for all project resource items and review results using the Material Planner Workbench.

Exercise steps

Part 1: Activate MRP from the MRP Mode form

1. Select **Master Explorer > Modules > Codes > Parameters > MRP Mode**. The **MRP Mode** form opens.
2. Select the **Use MRP Planning** check box.
3. Save the record.
4. Close the **MRP Mode** form.

Part 2: Open the MRP Planning form and run MRP planning

1. Select **Master Explorer > Modules > Material > Planning > Activities > MRP Planning**. The **MRP Planning** form opens with the **Regenerate** radio button selected.
2. Click **Plan**. A dialog box opens with the message, "Task Submitted."
3. Click **OK**.
4. Close the **MRP Planning** form.

Part 3: Generate MRP receipts

1. Select **Master Explorer > Modules > Material > Planning > Activities > Material Planner Workbench Generation**. The **Material Planner Workbench Generation** form displays.
2. Type **<three months from today>** in the **Ending Date** field.
3. Clear the **Buyer** field.
4. Leave all other fields as they defaulted.
5. Click **Process**. A dialog box opens with the message, "XXX MRP Receipt(s) were processed."
6. Click **OK**.
7. Close the **Material Planner Workbench Generation** form.

Part 4: Review information on the Material Planner Workbench form

1. Select **Master Explorer > Modules > Material > Planning > Material Planner Workbench**. The **Material Planner Workbench** form displays.
2. Verify **Purchase Order** is in the **View** field.
3. Select the **All Orders** radio button.
4. Locate the **GR-10000 Gravel, Grey** item in the grid.
5. Select the **Process Line** check box preceding **GR-10000, Gravel, Grey** in the **Item** list (this is the item with the required qty of 13,000). A dialog box opens with the message, "Process Vendor may not be [Blank] for Material Planner Workbench that has [Item: GR-10000] and [Due Date: mm/dd/yyyy]. Do you want to change the value now?"
6. Click **No**.
7. Scroll to the right to the **Vendor** column.
8. Click the **Vendor** field of your selected line.
9. Select **13-Marshland Realty Company** from the **Vendor** drop-down list. A dialog box opens with the message, "Every Material Planner Workbench where Item is GR-10000 and Due Date is mm/dd/yyyy or greater will have Vendor set to 13."
10. Click **Yes**.
11. Save the record.
12. Click **Planning Detail**. The **Planning Detail (Linked)** form opens.
13. Click each line on the grid and review the **Reference** field until you find one that matches your project (PROJ DX0000000X-30-1).
14. Review the **Projected On Hand** and **Outstanding Requirement** fields.
15. Close all forms.



Items with the type of Other or items having enough on-hand inventory do not display on the Material Planner Workbench.





Lesson 5: Project transactions

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- Define characteristics of project resource transactions.
- Describe project labor transactions.

Topics

- Project Parameters – Default tab
- Project Resource Transactions
- Project Labor Transactions
- Post Project Labor Transactions
- Employees – General tab
- Departments

Project Labor Transactions

The **Project Labor Transactions** activity is used to apply the cost of labor to a specific project or project task. The breakdown of the labor transaction can be viewed by cost code.

A project's actual labor costs are not updated until the labor transactions are posted.



The Project Rate is the Mfg Reg Rate found on the **Employee** form, Rates and Taxes tab.



Useful project reports

A useful report is the Unposted Labor Transactions Report. This report is accessed by selecting **Master Explorer > Modules > Project > Reports**.



Post Project Labor Transactions

Labor transactions are posted using the **Post Project Labor Transactions** activity. The cost of the labor is applied to the appropriate project account and can be viewed by cost code. The direct labor expense and applied overhead is credited to the appropriate department accounts.

By selecting "Print" and "Process" on the **Post Project Labor Transactions** form, labor transactions display that are ready for posting, depending on the selection criteria.

By selecting "Commit" and "Process," transactions post to the Project Control Ledger Distribution Journal.

Example journal entry

Below is an example of a labor transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Labor	360.96	
Project Account – OH	36.10	
Project Account – G & A	39.71	
Direct Labor Expense		100.96
Fixed Labor Overhead Applied		140.00
Variable Labor Overhead Applied		120.00
Project Parameter Account – Applied OH		36.10
Project Parameter Account – Applied G & A		39.71



Employees – General tab

The **Employees** form maintains employee data for projects, job scheduling and costs, Human Resources reports, inventory transactions, and payroll transactions.

The **Department** field on the **General** tab is key to the expense accounts used during labor posting. This is a required field on the form. The Department field specifies which department the employee is tied to for labor overhead calculations.

Labor costs can be tracked and posted against projects.

Access the Employees form by selecting **Master Employee > Modules > Employee > Employees**.



Departments

The **Departments** form defines specific areas of a company. The **Departments** form maintains labor overhead and general ledger account information.

Fixed and variable overhead rates are used to calculate overhead for an employee based on the employee's direct labor expense.

Account numbers are used to create journal entries when labor is posted to a project.

Access the Departments form by selecting **Master Explorer > Modules > Employee > HR > Organization > Departments**.



Project Resource Transactions

The cost and quantity of resources for a specific project can be issued to the project using the **Project Resource Transactions** activity. The cost code is required to post the transaction. The cost code for an item is initially accessed and assigned on the **Project Resources** form. Results of the posted resource transactions are viewed on the **Project Resources** form, **% Complete** tab.

Example journal entries

The tables below are examples of resource transactions for an inventoried and non-inventoried item as it affects the Project Control Ledger Distribution Journal.

Journal entry – inventoried item

The example below is an inventoried item as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Material	100.00	
Project Account – OH	10.00	
Project Account – G & A	11.00	
Inventory Material		100.00
Project Parameter Account – Applied OH		10.00
Project Parameter Account – Applied G & A		11.00

Journal entry – non-inventoried item

The example below is a non-inventoried item as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Material	200.00	
Project Account – OH	20.00	
Project Account – G & A	22.00	
Expense Account		200.00

	Debit	Credit
Project Parameter Account – Applied OH		20.00
Project Parameter Account – Applied G & A		22.00



A useful report is the Project Transactions Report. This report is accessed by selecting **Master Explorer > Modules > Project > Reports**.





Exercise 5.1: Enter labor transactions and issue resources

In this exercise, you will enter labor transactions for the employees who worked on surveying and excavation and have turned in their time sheets and issue resources.

Exercise steps

Part 1: Enter labor transactions on the Project Labor Transactions form

1. Select **Master Explorer > Modules > Project > Activities > Project Labor Transactions**. The **Project Labor Transactions (Filter in Place)** form opens.
2. Run **Filter In Place**.
3. Add a new record.
4. Select **4-Taylor, Kirk C.** from the **Employee** drop-down list.
5. Select **<your project number>** from the **Project** drop-down list.
6. Select **10-Survey the site** from the **Task** drop-down list.
7. Select **GL-General Labor** from the **Cost Code** drop-down list.
8. Type **20** in the **Total Hours** field.
9. Save the record.

10. Add the labor transactions for your project, one at a time, following steps 3-9 using the table below.

Employee Name	Task	Cost Code	Total Hours (of labor)
5 – Hathaway, Susan V.	10 – Survey the site	GL-General Labor	20 hours
15 – Wong, Jerry A.	20 – Excavate the site	GL-General Labor	20 hours
8 – King, Brenda A.	20 – Excavate the site	GL-General Labor	20 hours
4 – Taylor, Kirk C.	20 – Excavate the site	GL-General Labor	20 hours

11. Save the records.
12. Close the **Project Labor Transactions** form.

Part 2: Post labor transactions on the Post Project Labor Transactions form

1. Select **Master Explorer > Modules > Project > Activities > Post Project Labor Transactions**. The **Post Project Labor Transactions** form opens.
2. Select **<your project number>** from the **Project** drop-down list for the **Starting** field.
Note: The **Ending** field assumes the value you entered in the **Starting** field.
3. Verify the **Print** radio button is selected.
4. Click **Process**. The report viewer begins to open; you can close it. A dialog box opens with the message, “5 Project Labor Transaction(s) were processed.”
5. Click **OK**. A dialog box opens with the message, “Report Submitted.”
6. Click **OK** again.
7. Select the **Commit** radio button.
8. Click **Process**. A dialog box opens with the message, “[Post Project Labor Transactions] will be performed.”
9. Click **OK**. A dialog box opens with the message, “5 Project Labor Transaction(s) were posted.”
10. Click **OK** again.
11. Close the **Post Project Labor Transactions** form.

Part 3: Issue resources on the Project Resource Transaction form



You would normally have on hand quantities and these resources do not have on hand quantities so the exercise is built to issue what is required and let inventory go negative.

1. Select **Master Explorer > Modules > Project > Activities > Project Resource Transactions**. The **Project Resource Transactions** form displays.
2. Select **<your project number>** from the **Project** drop-down list.
3. Select **30-Construct** the track from the field to the right of the **Project** drop-down list.
4. Select **GR-10000-Gravel, Grey-1** from the **Item** drop-down list.
5. Type **13000** in the **Qty** field.
6. Tab out of the field. A dialog box opens with the message, "On Hand is 0.0000000 and will be - 13000.0000000 for Item Stockroom Location that has [Whse: MAIN] and [Item: GR-100000] and [Location: STOCK]."
7. Click **OK**.
8. Click **Process**. A dialog box opens with the message, "On Hand is 0.0000000 and will be - 13000.0000000 for Item Stockroom Location that has [Whse: MAIN] and [Item: GR-100000] and [Location: STOCK]."
9. Click **OK**. A dialog box opens with the message, "[Project Resource Transactions] was successful."
10. Click **OK**.
11. Verify the **Qty Issued** and **Required** quantities are the same.
12. Issue resources, following steps 4-11 for the following:

Item	Description	Resource Line Number	Quantity
TS-11000	Soil, Top	2	500,000
SA-10000	Sand	3	10,000
CB-10000	Block, Cinder	4	5,000
RT-10000	Ties, Railroad	5	1,000
RB-10000	Ramp, Bicycle	6	3
WP-10000	Pond, Water Hazard	7	2

13. Close the **Project Resource Transactions** form.

Part 4: View project costs on the Projects form

1. Return to the **Projects** form.
2. Refresh the current collection.
3. Click the **% Complete** tab.

4. View **Actual Costs and Forecast Costs**.
5. Click **Tasks**. The **Project Tasks (Linked)** form displays.
6. Click the % **Complete** tab.
7. View **Actual Costs** for Task 10 and Task 20.
8. Close the **Project Tasks (Linked)** form.
9. The **Projects** form remains open.

Check your understanding



Where can you view the results of a posted project resource transaction?





Lesson 6: Purchasing and projects

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- Describe the benefit of cross-referencing purchase orders to a project.
- Explain how to create purchase orders cross-referenced to a project.

Topics

- Project Resources – General tab, cross-reference
- Purchase Order Receiving

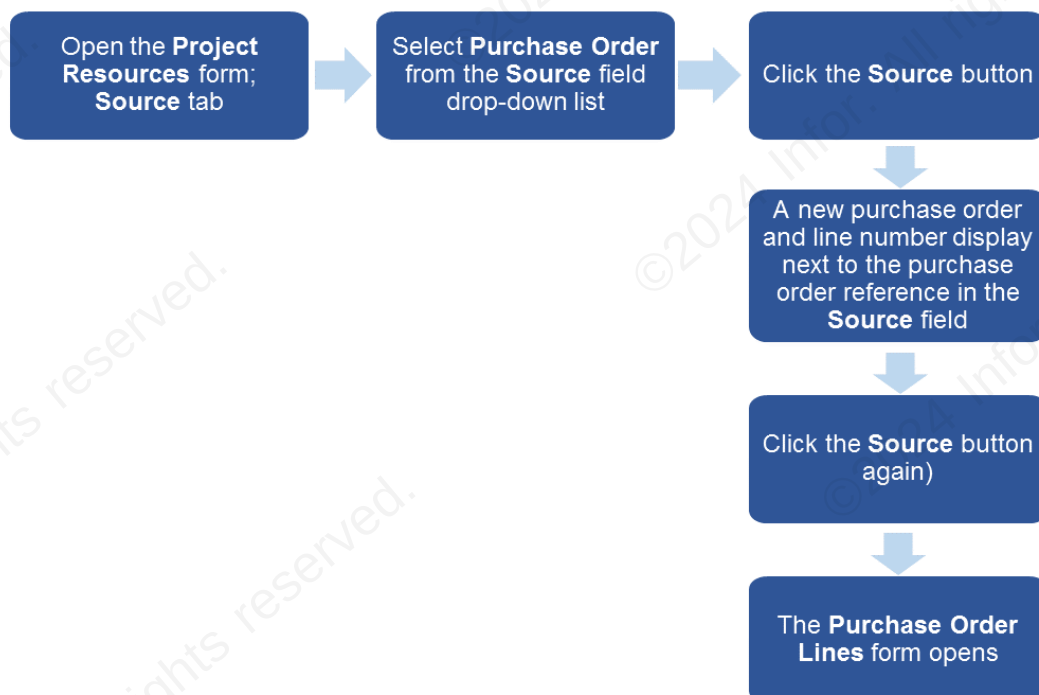
Project Resources – Source tab, cross-reference

Purchase orders can be cross-referenced to a project from the **Project Resources** form, **Source** tab. This allows the ability to capture the purchase order cost on the project.



If the resource item is in the items table, the value in its Cost Type field must be Actual for the actual purchase order cost to flow to the project. When you do the PO receipt the system automatically issues the resource to the project. When that occurs, the item is received into inventory and then issued to the project. Therefore, the best way of assuring that the PO cost is posted to the project is for the item to have a cost method of last in, first out (LIFO). Otherwise, the cost at which it is received at may not be the cost at which it is issued to the project. If the item's cost Type is Standard, it is issued to the project at its standard unit cost.

Follow these steps to cross-reference a project resource to a purchase order:



Cross reference to a PO

Purchase Order Receiving

Materials can be received through a purchase order that is referenced to a project and project task.



Useful project reports

A useful report is the Project Transactions Report. This report is accessed by selecting **Master Explorer > Modules > Project > Reports**.

Example journal entries

Examples of journal entries for items maintained and not maintained in Item Maintenance are displayed in the following tables.

Inventoried item receiving transaction – Purchasing Ledger Distribution Journal

Below is an example of a receiving transaction for an inventoried item as it affects the Purchasing Ledger Distribution Journal:

	Debit	Credit
Inventory Material	100.00	
Vouchers Payable		100.00

Inventoried item receiving transaction – Project Control Ledger Distribution Journal

Below is the same receiving transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Material	100.00	
Project Account – OH	10.00	
Project Account – G & A	11.00	
Inventory Material		100.00
Project Parameter Account – Applied OH		10.00
Project Parameter Account – Applied G & A		11.00

Non-inventoried item receiving transaction – Purchasing Ledger Distribution Journal

Below is an example of a receiving transaction for a non-inventoried item as it affects the Purchasing Ledger Distribution Journal:

	Debit	Credit
Expense Account	16.03	
Vouchers Payable		16.03

Non-inventoried item receiving transaction – Project Control Ledger Distribution Journal

Below is the same receiving transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Material	16.03	
Project Account – OH	1.60	
Project Account – G & A	1.76	
Expense Account		16.03
Project Parameter Account – Applied OH		1.60
Project Parameter Account – Applied G & A		1.76



Exercise 6.1: Purchase and receive resources

In this exercise, you will purchase and receive resources.

Exercise steps

Since construction has begun on the bicycle racing track, it is time to purchase the resources that will be needed for the landscaping task.

Part 1: Start from the Project Resources form and purchase resources

1. Select **Master Explorer > Modules > Project > Project Resources**. The **Project Resources (Filter In Place)** form opens.
2. Select **<your project number>** from the **Project** drop-down list.

3. Select **40-Landscaping and adding finishing touches** from the **Task** drop-down list.
4. Run **Filter In Place**. The three resources associated with the task display in grid view.
5. Select the **first resource (Trees)** from the grid view.
6. Click the **Source** tab.
7. Verify **Purchase Order** is in the **Source** field.
8. Click **Source**. A dialog box opens with the message, "[Cross Reference] will be performed from Project Resource to Purchase Order."
9. Click **OK**.
10. Review the unlabeled reference to the purchase order (DP00000XXX) and line number (1).
11. Write down the **Purchase Order** number: _____.
12. Select the **second resource (Shrubs)** from the grid view.
13. Type **<your PO number>** in the **Purchase Order** field below the **Source** field.
Note: This will combine the attached resources into one purchase order.
14. Save the record.
15. Click **Source**. A dialog box opens with the message, "[Cross Reference] will be performed from Project Resource to Purchase Order."
16. Click **OK**.
17. Follow steps **12-16** for the third resource (Sod).
18. Click the **Source** button again. The **Purchase Order Lines** form opens.
19. Click **PO**. The **Purchase Orders (Linked)** form opens.
20. Select **13-Marshland Realty Company** from the **Vendor** drop-down list. A dialog box opens with the message, "Vendor has changed. You may need to re-calculate cost."
21. Click **OK**.
22. Select **Ordered** from the **Status** drop-down list. A dialog box opens with the message, "Vendor has changed. You may need to re-calculate cost. [Change PO Status] will be performed."
23. Click **OK**.
24. Save the record.
25. Close the **Purchase Orders (Linked)** form. The **Purchase Order Lines** form displays again.
26. Run **Filter In Place**.
27. Type **<your PO number>** in the **PO** field.
28. Run **Filter In Place**. Three resources display on the grid.
29. Select the **first PO line** from the grid view.
30. Select **Ordered** from the **Status** drop-down list. A dialog box opens with the message, "[Change PO Line/Release Status] will be performed."
31. Click **OK**.
32. Save the record.

33. Follow steps **30-32** for the remaining two PO lines one at a time.
34. Close the **Purchase Order Lines** form. Leave the **Project Resources** form open as you will use it later.

Part 2: Receive resources on the Purchase Order Receiving form

Note: It is four weeks later and time to receive the materials from the purchase order.

1. Select **Master Explorer > Modules > Vendor > Purchase Orders > Activities > Purchase Order Receiving**. The **Purchase Order Receiving** form opens.
2. Select **<your PO number>** from the **PO Number** drop-down list. The three purchase order lines display in the grid in the middle of the form and action buttons display above the grid.
3. Click **Select All**. All three lines check boxes are activated.
4. Click **Receive**. A dialog box opens with the message, "[Project Resource Transactions] will be performed. Ref Type is K **<your project number>** 40 1 for PO Line/Release that has [PO: DP00000XXX] and [Line: 1] and [Release: 0]."
5. Click **OK**. A dialog box opens with the message, "[Project Resource Issue] was successful. "[Project Resource Transactions] will be performed. Ref Type is K **<your project number>** 40 2 for PO Line/Release that has [PO: DP00000XXX] and [Line: 2] and [Release: 0]."
6. Click **OK**.

Note: The dialog box appears one more time referencing each PO line.

7. Click **OK** after each message. After, a dialog box opens with the message, "[Project Resource Issue] was successful."
8. Click **OK**. A dialog box opens with the message, "3 Pending PO Receiving Transaction(s) were posted."
9. Click **OK**.
10. Close the **Purchase Order Receiving** form.

Part 3: View project costs on the Project Resources form

1. Return to the **Project Resources** form.
2. Refresh the current collection.
3. Select the **first resource (Trees)** from the grid view.
4. Click the **Source** tab.

Note: The order line status is now Filled and the quantity for the line has been received.

5. Click the **% Complete** tab.
6. View the **Actual Cost** posted to the project.
7. View the **Actual Cost** for the remaining two resources, following steps **3-5**. The **Projects Resources** form remains open.



The **Projects** form provides a cumulative view of all cost codes. You can also view costs broken down by cost code on the **Project Tasks** form.

Check your understanding



What is the importance of cross-referencing a purchase order to a project?





What journals are affected by cross-referencing a receiving transaction?





Lesson 7: Job orders and projects

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- Describe aspects of jobs orders cross-referenced to a project.

Topics

- Project Resources – General tab
- Job Orders
- Job Material Transactions
- Unposted Job Transactions
- Post Job Transactions

Project Resources – Source tab

A project resource can be cross-referenced from the **Project Resources** form, **Source** tab to the Job Orders form. This allows job order costs to be tracked for a project.

Once the form is opened, select Job from the Source drop-down list, click **Source**. A new Job Order number displays next to the Source field. Clicking **Source** again opens the **Job Orders** form.



When a job is cross-referenced to a project, it loses its cost type/cost method visibility since a job will never move into inventory at completion and all costs remain at the project level.



Job Orders

The **Job Orders** form contains information cross-referenced from the **Project Resources** form.

Scheduling tab

The **Scheduling** tab references:

- Item
- Item Description
- Released

References tab

The **References** tab references:

- Project
- Project Task
- Customer
- Due Date



When performing Project Cost Roll Up, the cost codes found on the **Work Centers** form, Costing tab and the **Product Codes** form, Project tab are used for planning the cost to manufacture the job order linked to the project.



Exercise 7.1: Cross-reference a resource to a new job order

In this exercise, you will cross-reference a resource to a new job order.

The project is progressing nicely. However, there is still work to be done. You need to prepare to manufacture the bicycle that will be used for the last project task, which is track testing.

Cross-reference the project resource for the FA-30000 to a new job order.

Exercise steps

1. Return to the **Project Resources** form.
2. Run **Filter In Place**.
3. Select **50-Test the track** from the **Task** drop-down list.
4. Run **Filter In Place**. The resource associated with the task displays in grid view.
5. Click the **Source** tab.
6. Verify **Job** is selected in the **Source** field.
7. Click the **Source** button. A dialog box opens with the message, “[Cross Reference] will be performed from Project Resource to JOB.”

8. Click **OK**. Notice the reference to the job order below the **Source** field.
9. Write down the **Job** number: _____.
10. Click the **Source** button. The **Job Orders** form opens and **MB-10000** displays in the **Item** field.
11. Click the **References** tab.
12. Review the references to the **Project**, **Project Task**, and **Customer**.
13. Click the **Schedule** tab.
14. Select **Released** from the **Status** field.
15. Save the record. A dialog box opens with the message, "[Copy Current BOM] will be performed."
16. Click **OK**.
17. Close the **Job Orders** form. You are returned to the **Project Resources** form.

Job Material Transactions

Job materials referenced to a project can have the material costs capitalized on the project. A cost code must be specified when performing a job material transaction. You can view the project to see the actual costs accumulated by cost code.



The net WIP for a job order referenced to a project should always be zero in the distribution journals.



Useful project report

A useful report is the Project Transactions Report. This report is accessed by selecting **Master Explorer > Modules > Project > Reports**.

Example journal transactions

Examples of journal entries for job material transactions are displayed in the following tables.

Inventoried item job material transaction – Shop Floor Ledger Distribution Journal

Below is an example of a material transaction for a system inventoried item as it affects the Shop Floor Ledger Distribution Journal:

	Debit	Credit
Work in Process	500.00	
Inventory		500.00

Inventoried item job material transaction – Project Control Ledger Distribution Journal

Below is the same material transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Material	500.00	
Project Account – OH	50.00	
Project Account – G & A	55.00	
Work in Process		500.00

	Debit	Credit
Project Parameter Account – Applied OH		50.00
Project Parameter Account – Applied G & A		55.00

Non-inventoried item job material transaction – Shop Floor Ledger Distribution Journal

Below is an example of a material transaction for a system non-inventoried item as it affects the Shop Floor Ledger Distribution Journal:

	Debit	Credit
Work in Process	20.00	
User Specified Asset Account		20.00

Non-inventoried item job material transaction – Project Control Ledger Distribution Journal

Below is the same material transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Material	20.00	
Project Account – OH	2.00	
Project Account – G & A	2.20	
Work in Process		20.00
Project Parameter Account – Applied OH		2.00
Project Parameter Account – Applied G & A		2.20



Unposted Job Transactions

The **Unposted Job Transaction** form is used to record job order labor and move transactions. If a job order is cross-referenced to a project, labor and machine overhead cost will be accumulated on the project.

A cost code is required for the transaction to be posted if the job order is linked to a project.



Post Job Transactions

The **Post Job Transactions** form is used to post jobs so that job costs are recognized and the costs are accumulated to the project.



Useful project report

A useful report is the Project Transactions Report. This report is accessed by selecting **Master Explorer > Modules > Project > Reports**.

Example journal transactions

Examples of journal entries for post job transactions are displayed in the following tables.

Labor transaction – Shop Floor Ledger Distribution Journal

Below is an example of a labor transaction as it affects the Shop Floor Ledger Distribution Journal:

	Debit	Credit
WIP Fixed Overhead	20.00	
WIP Variable Overhead	20.00	
WIP Labor	30.00	
Direct Labor Expense		30.00
Fixed Labor Overhead Applied		20.00
Variable Labor Overhead Applied		20.00

Labor transaction – Project Control Ledger Distribution Journal

Below is the same labor transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Labor	70.00	
Project Account – OH	7.00	
Project Account – Labor	7.70	

	Debit	Credit
WIP Labor		30.00
WIP Fixed Overhead		20.00
WIP Variable Overhead		20.00
Project Parameter Account – Applied OH		7.00
Project Parameter Account – Applied G & A		7.70



There are no inventory transactions when you complete items since all the costs stay at the project level and do not go into inventory.





Exercise 7.2: Add and post move transactions

In this exercise, you will add and post move transactions.

Exercise steps

Part 1: Add move transactions using the Unposted Job Transactions form

1. Select **Master Explorer > Modules > Production > Jobs > Activities > Unposted Job Transactions**. The **Unposted Job Transactions (Filter in Place)** form opens.
2. Run **Filter In Place**.
3. Add a new record.
4. Select **Move** from the **Transaction Type** drop-down list.
5. Select **<your job number> Released MB-10000** from the **Job** drop-down list.
6. Select **20** from the **Operation** drop-down list.

7. Type 5 in the **Completed** field.
Note: A quantity of 5 automatically populates in the **Moved** field.
8. Select the **Oper Complete** check box.
9. Save the record.
10. Add a new record.
11. Verify **Move** is selected in the **Transaction Type** field.
12. Verify **<your job number> Released MB-10000** is selected in the **Job** field.
13. Verify **40** is selected in the **Operation** field.
14. Type 5 in the **Completed** field.
15. Tab out of the field. A dialog box opens with the message, "Completed is greater than Qty Required for Job Operation that has [Oper: 40]. Qty Required is 0.00000000."
16. Click **OK**.
17. Select the **Oper Complete** check box.
18. Save the record. A dialog box opens with the message, "Complete? is equal to No for Job Operation that has [Oper: 20]."
19. Click **OK**.
20. Click **Post Job Transactions**. The **Post Job Transactions** form opens.
21. Select the **Post "Complete" Transactions** check box.
22. Select **<your job number>** from the **Job** drop-down list for both **Starting** field.
Note: The **Ending** field assumes the value you entered in the **Starting** field.
23. Click **Process**. A dialog box opens with the message, "[Post Job Transactions] will be performed."
24. Click **OK**. A dialog box opens with the message, "[Post Job Transaction(s)] was successful."
25. Click **OK**.
26. Close the **Post Job Transactions** form and the **Unposted Job Transactions** form.

Part 2: View project costs

1. Return to the **Projects** form.
2. Refresh the current collection.
3. Click the **% Complete** tab.
4. Select the **Job Cost Codes** radio button.
5. View the **Actual Cost** column and totals.
Notice that each job is broken down by Cost Code.
6. Leave the **Projects** form open.

Check your understanding



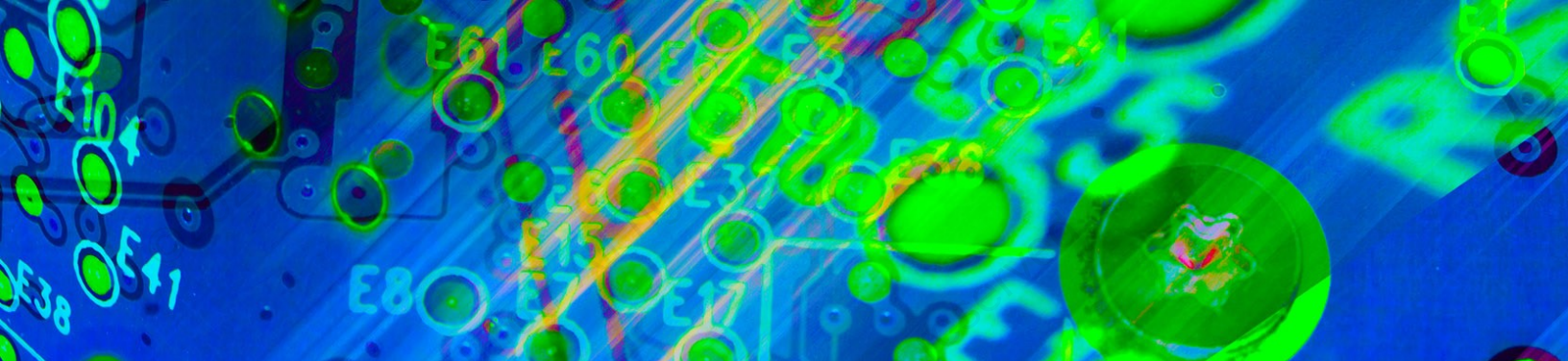
Why does a job lose its cost type/cost method visibility when referenced to a project?





What should the net WIP for a job order referenced to a project be in the distribution journals?





Lesson 8: Vouchering and projects

Estimated time

.25 hours

Learning objectives

In this lesson, you will:

- Describe the purpose of using vouchers/adjustments for projects.
- List details required when entering a voucher adjustment distribution record.

Topics

- A/P Vouchers and Adjustments
- Voucher Adjustment Distribution
- A/P Voucher Posting

A/P Vouchers and Adjustments

The **A/P Vouchers and Adjustments** form is used to capture miscellaneous expenses that are accumulated on a project. A voucher/adjustment can be created manually. The voucher/adjustment can be linked to a project and project task via the **Voucher Adjustment Distribution** form.



Voucher Adjustment Distribution

The project number, task number, and cost code must be entered when creating a voucher adjustment distribution record. The cost code determines which project account is used in the journal entry.

If applicable, the project will have overhead and/or G & A overhead calculated.



A/P Voucher Posting

After posting the voucher/adjustment, the only entry found in the Project Control Distribution Journal is overhead and/or G & A overhead based on the voucher/adjustment amount.

Example journal transactions

The tables below are examples of accounts payable transactions.

Accounts payable transaction – Accounts Payable Ledger Distribution Journal

Below is an example of an accounts payable transaction as it affects the Accounts Payable Ledger Distribution Journal:

	Debit	Credit
Project Account - Other	100.00	
Accounts Payable		100.00

Accounts payable transaction – Project Control Ledger Distribution Journal

Below is the same accounts payable transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
Project Account – Overhead	10.00	
Project Account – G & A	11.00	
Project Parameter Account – Applied OH		10.00
Project Parameter Account – Applied G & A		11.00





Exercise 8.1: Record project expense using a voucher

In this exercise, you will link a voucher to the project which records the cost of the expense.

A voucher must be created to set up payment for some miscellaneous travel expenses that were incurred when the supervisor traveled from the West Coast to inspect the construction project task progression. Jay Richards' total travel expenses were calculated to be \$1052.30.

Exercise steps

Part 1: Create the voucher on the A/P Vouchers and Adjustments form

1. Select **Master Explorer > Modules > Finance > Accounts Payable > A/P Vouchers and Adjustments**. The **A/P Vouchers and Adjustments (Filter In Place)** form opens.
2. Run **Filter In Place**.
3. Add a new record.
4. Select **16-Riggs, Joshua A.** from the **Vendor** drop-down list.
5. Type **<current month and current year> Expenses** in the **Invoice** field.
6. Type **1052.30** in the **Purch Amt** field.
7. Save the record.
8. Write down your voucher number: _____.

Part 2: Add a distribution to the project and project task for account number 58600

The cost code associated with travel expenses is MCE – Miscellaneous Expenses. Since the original forecast did not include a budgeted amount for miscellaneous expenses, it is assumed that the travel costs will be absorbed in overhead calculations.

1. Click **Distribution**. The **Voucher Adjustment Distribution (Linked)** form opens.
2. Select **<your project number>** from the **Project** drop-down list.
3. Select **30-Construct the track** from the task number drop-down list.
Note: This drop-down list is not labeled. It is located to the right of the **Project** drop-down list.
4. Select **MCE-Miscellaneous Expenses** from the cost code drop-down list.
Note: This drop-down list is not labeled. It is the second field to the right of the **Project** drop-down list.
5. Type **1052.30** in the **Amount** field.
6. Select **58600-Project Control - Other** from the **Account** drop-down list.
7. Save the record.
8. Close the **Voucher Adjustment Distribution** form. You are returned to the **A/P Vouchers and Adjustments** form.

Part 3: Post the voucher to capture miscellaneous expenses on the project

1. Return to the **A/P Vouchers and Adjustments** form.

2. Click **A/P Voucher Posting**. The **A/P Voucher Posting** form displays. **Note:** The **Print** radio button is selected.
3. Click **Process**. A dialog box opens with the message, "[AP Voucher Posting] will be printed."
4. Click **OK**. The report viewer begins to open; you can close it. A dialog box opens with the message, "Report Submitted."
5. Click **OK**.
6. Select the **Commit** radio button.
7. Click **Process**. A dialog box opens with the message, "[Post] will be performed."
8. Click **OK**. A dialog box opens with the message, "1 A/P Transaction(s) were posted."
9. Click **OK**.
10. Close the **A/P Voucher Posting** form.
11. Close the **A/P Vouchers and Adjustments** form.

Part 4: Return to the Projects form and view project costs

1. Return to the **Projects** form.
2. Refresh the current collection.
3. Click the **% Complete** tab.
4. Click the **Project Cost Codes** radio button.
5. View the **Cost Code** column.

Notice how the new **Cost Code-MCE** is captured for the project and the actual amount equals the expense entered for Joshua Riggs.

Check your understanding



What journal entry affects the Project Control Ledger Distribution Journal after posting a voucher/adjustment?





Lesson 9: Shipping and projects

Estimated time

.25 hours

Learning objectives

In this lesson, you will:

- Discuss how to ship product directly from a project.
- Identify ways to create advance ship notices for a project.

Topics

- Project Resource Shipping
- Project Resources – Shipping tab
- Project Packing Slip
- Project Advance Ship Notices

Project Resource Shipping

The **Project Resource Shipping** form allows you to ship product directly from the project rather than the customer order.

All resources display in the grid with the quantities issued to the project as well as those shipped to the project site. The application automatically populates the **Ship Date** and **Qty to Ship** fields. The **Qty to Ship** field displays the difference between what has been issued to the project and what has been shipped.

Individual line items or the entire batch of project resources can be shipped.

To return inventory to stock (before invoicing has occurred), perform a regular shipping transaction, entering a negative number for the item or items being returned. The inventory account (asset) will be debited and the inventory quantity will be increased.

The detail transactions that are generated will be stored with the user ID and the date of the posting and can be viewed from the **Shipping** tab of the **Project Resources** form.



Since the cost related to these resources has already been assigned to WIP, there are no distribution journal transactions created when performing a shipment. This process simply lets you know whether the materials are at their site or at the customer's site.



Project Resources – Shipping tab

The **Project Resources** form, **Shipping** tab displays the detail transactions that have processed against the project using the **Project Resource Shipping** activity.

The **Sequence Number**, **Ship Date**, **Quantity Shipped** (or **Returned**), **Transaction Date**, and **User Name** (who performed the posting process) fields display in the grid.

Two fields at the bottom of the form show the Total Packed and the Total Shipped for the resource.



Project Packing Slip

A Project Packing Slip can be printed for the project resources being shipped. Select the appropriate criteria on the **Project Packing Slip** form and print the report.



The **Project Pre-Ship Packing Slip** form can also be used.



Project Advance Ship Notices

The **Project Advance Ship Notices** form allows you to create a Bill of Lading for the project resources that are being shipped. The **Project Advance Ship Notices** form can be either manually created or generated one of two ways:

- Generate from the project; this allows you to generate a Project Advance Ship Notice based on the project resources
- Generate from the packing slip; this allows you to generate a Project Advance Ship Notice based on the project packing slip

The Recalculate option allows you to recalculate the total charges for the Project Advance Ship Notices.

Access the **Project Advance Ship Notices** form by selecting **Master Explorer > Modules > Project > Utilities > Project Advance Ship Notices**.



Useful project reports

Useful reports are the Reprint Project Packing Slip Report and the Print Project Advance Ship Notice Report. Both reports are accessed by selecting **Master Explorer > Modules > Project > Reports**.



Exercise 9.1: Ship resources to the customer

In this exercise, you will ship resources to the customer.

Since much of the construction work has started on the bicycle track project, it is time to ship many of the resources to the customer so construction can finish.

Exercise steps

Part 1: Ship all resources issued to the project directly to the customer associated with Task 30

1. Select **Master Explorer > Modules > Project > Activities > Project Resource Shipping**. The **Project Resource Shipping (Filter In Place)** form opens.
2. Select **<your project number>** from the **Project** drop-down list.
3. Run **Filter In Place**.
4. Select **Task 30-Construct the track** from the grid view. Resources associated with **Task 30** display in the grid.
5. View the values in the **Qty To Ship** column.
6. Click **Ship Batch**. A dialog box opens with the message, "[Batch Resource Shipment] will be performed for Project Resource that has [Project: DX0000000X] and [Task #:30]."
7. Click **OK**. A dialog box opens with the message, "[Project Resource Shipment] was successful."
8. Click **OK**.

Note: The values moved from the **Qty To Ship** column to **Qty Shipped** column.

9. Close the **Project Resource Shipping** form.

Part 2: Print Project Packing Slip for all resources associated with Task 30

1. Select **Master Explorer > Modules > Project > Activities > Project Packing Slip**. The **Project Packing Slip** form opens.
2. Select **<your project number>** in the **Project** drop-down list for the **Starting** field.
Note: The **Ending** field assumes the value you entered in the **Starting** field.
3. Select **30-Construct the track** from the **Task/Resource** drop-down list.
4. Click **Process**. A **Project Packing Slip Selection (Modal)** form opens showing all lines to be shipped for **Task 30**. You can add **Weight, # Packages** and **Ship Via** information.
5. Click **Print**. The report viewer begins to open; you can close it. A dialog box opens with the message, "Report Submitted."
6. Click **OK**.
7. Close the **Project Packing Slip** form.

Part 3: View the report

1. Select **Master Explorer > Modules > System > Report Output Files**. The **Report Output Files (Filter In Place)** form opens.
2. Run **Filter In Place**.
3. Select **ProjectPackingSlipReportViewer** from the grid.
4. Click **View**. A dialog box opens with the message, "Do you want to open or save ...?".
5. Click **Open**. The **Report Viewer** form displays.
6. Review the packing slip.
7. Close the **Report Viewer** form.
8. Close the **Report Output Files** form. You are returned to the **Projects** form.

Part 4: View details of the shipping

1. Click **Tasks**.
2. Select the line associated with Task 30.
3. Click **Resources**. The **Project Resources (Linked)** form opens.
The seven resources associated with Task 30 display in the grid view.
4. Click the **Shipping** tab.
5. Select the first resource from the grid view.
6. Note the following fields:
 - Ship Date
 - Shipped
 - Transaction Date
 - User Name
 - Total Packed (displays at the bottom of the form)
 - Total Shipped (displays at the bottom of the form)
 - View the remaining resources, following steps 6-8.

7. Close the **Project Resources (Linked)** form.
8. Close the **Project Tasks (linked)** form. You are returned to the **Projects** form.

Check your understanding



Are journal entries created when shipping project resources from a project? Why or why not?





Lesson 10: Work in process (WIP) and projects

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- List the three WIP Relief Method options.
- Identify product code requirements for WIP Distribution Accounts.
- Explain how to alter the WIP Relief Method for individual project tasks.
- State how to print/post project WIP relief.

Topics

- Project Parameters – General tab
- Distribution Accounts maintenance
- Projects - WIP
- Project Tasks - WIP
- Print/Post Project WIP Relief
- Project WIP Adjustments

Project Parameters – General tab

The WIP Relief Method specified on the **Project Parameters** form, **General** tab will be the default for the WIP Relief Method. The three options are:

- Manual
- Shipment
- Revenue Recognition



Distribution Accounts maintenance

Product codes for distribution accounts must be entered for cost of goods sold (COGS) transactions. These accounts are used for posting project WIP dollars to COGS.

Access the **Distribution Accounts** form by selecting **Master Explorer > Modules > Material > Inventory > Distribution Accounts**.



Projects – WIP

The **Projects** form allows you to enter a project that can be set up to calculate the forecast cost for completing the project. The project can be cross-referenced from the **Customer Order Lines** form or copied from an estimate project. Projects can also be manually created.

Projects – WIP tab

The **Projects** form, **WIP** tab displays the WIP Accounts for the project. These values are broken down into the five WIP cost accounts:

- Material
- Labor
- Other
- Overhead
- G & A

Projects – WIP Relief tab

The **Projects** form, **WIP Relief** tab displays the **WIP Relief Method**, **Start WIP Relief** date field, and **Project WIP Relief Accounts**. WIP Relief is moving costs from the project WIP Relief Accounts to COGS accounts when a customer order line is cross-referenced to a project (job).

WIP Relief Method

The **WIP Relief Method** drop-down list displays on the **Projects** form. **Note:** The value in this field retrieves the value defined within the **Projects Parameters** form, **General** tab, **WIP Relief Method** field.

This field can be altered to reflect the best WIP Relief Method for each project task.

The options found in the **WIP Relief Method** drop-down list are as follows:

WIP Relief Method	Description
Manual	The Manual WIP Relief Method expects that you enter a Start Date on the Projects form, WIP Relief tab. As soon as this date is entered, WIP relief can occur during processing of the Print/Post Project WIP Relief activity.
Shipment	The Shipment WIP Relief Method watches for the shipment of a customer order line item that is cross-referenced to a project. Once the shipment occurs, a Start Date is entered on the Projects form, WIP Relief tab. When the Print/Post Project WIP Relief activity is run, the project is triggered to have the WIP relieved to the appropriate COGS accounts.
Revenue Recognition	The Revenue Recognition WIP Relief Method is used to relieve WIP when revenue recognition takes place for the project. Revenue is recognized when the Print/Post

WIP Relief Method	Description
	Revenue Milestone activity is processed. WIP can be relieved during the processing of this activity.



When the WIP Relief Method is Revenue Recognition, WIP is only relieved at the project level.

The WIP Relief accounts on the **WIP Relief** tab of the **Projects** form default from the **COGS** tab of the **Distribution Accounts** form where the Product Code on the Project matches the Product Code on the Distribution Accounts form. The COGS accounts from the **Distribution Accounts** form are mapped as follows:

Project accounts	Is mapped to ...	COGS accounts
Material		Material
Labor		Labor
Other		Outside Services
Overhead		Variable Overhead
G & A		Fixed Overhead

Projects – Cost tab

The **Projects** form, **Cost** tab displays all accumulated costs and WIP Relieved costs for the project. These values are broken down into the five WIP cost accounts:

- Material
- Labor
- Other
- Overhead
- G & A

Total WIP Relieved and Net WIP values also display on this form.

Net WIP = Total Cost – Total WIP Relieved



The Cost to Complete Snap Shot is a good way to keep a record of WIP costs for a project.



Project Tasks – WIP

Project Tasks – WIP tab

The **Project Tasks** form, WIP tab shows all the accounts numbers based on the product code from the General tab of the Projects form. These values are broken down into the following five WIP cost accounts for projects:

- Material
- Labor
- Other
- Overhead
- G & A



Project Tasks – WIP Relieved tab

WIP is only relieved at the project task level when using the WIP Relief Method of “Manual” or “Shipment” found on the **Projects** form, **WIP Relief** tab.

The WIP Relief accounts listed default from the **Product Code for Distribution Accounts** form displaying the COGS accounts. The COGS accounts from the **Distribution Accounts** form are mapped the same for tasks as they are at the project level.

The COGS accounts from the **Distribution Accounts** form are mapped as follows:

Project account	Is mapped to ...	COGS account
Material		Material
Labor		Labor
Other		Outside Services
Overhead		Variable Overhead
G & A		Fixed Overhead

Project Tasks – Costs tab

- The **Project Tasks** form, **Cost** tab displays all accumulated costs and WIP Relieved costs for the project task. These values are broken down into the five WIP cost accounts:
- Material
- Labor
- Other
- Overhead
- G & A

Total WIP Relieved and Net WIP also display for the project task.

Net WIP = Total Cost – Total WIP Relieved



Print/Post Project WIP Relief

The **Print/Post Project WIP Relief** activity moves unrelieved cost amounts out of WIP costs to the COGS accounts.

The following table describes two sections on the **Print/Post Project WIP Relief** activity:

Section	Description
Print Level	The Print Level section allows you to view a WIP Relief report summarized per project or task detail.
Post Level	The Post Level section allows you to post WIP Relief as follows: • No posting – Runs the report and will not post any activity • Project – Posts WIP Relief at the project(s) summarized level to the Project Control Ledger Distribution Journal • Task – Posts WIP relief in project task detail to the Project Control Ledger Distribution Journal



If the WIP Relief Method is Revenue Recognition, the Post/Print Project WIP Relief activity does not need to run. WIP can be relieved when revenue is recognized.

Example journal transaction

Below is an example of a WIP Relief Posting transaction as it affects the Project Control Ledger Distribution Journal:

	Debit	Credit
COGS – Material	100.00	
COGS – Labor	50.00	
Project Account – Material		100.00
Project Account - Labor		50.00



Exercise 10.1: Relieve WIP using the Manual WIP Relief Method

In this exercise, you will relieve WIP for Project DX00000004, so that you can use your project for Revenue Recognition and Invoicing.

Exercise steps

Part 1: Verify WIP Relief Method is Manual

1. Return to the **Projects** form.
2. Run **Filter In Place**.
3. Type **<DX00000004>** in the **Project** field.
4. Run **Filter In Place**.
5. Click the **WIP Relief** tab.
6. Select **Manual** from the **WIP Relief Method** field.
7. Type **<today's date>** in the **Start Date** field to the right of the **WIP Relief Method** field.
8. Save the record.

Part 2: Roll up costs on the project

1. Select **Master Explorer > Modules > Project > Utilities > Roll Up Project Cost**. The **Roll Up Project Cost** form opens.
2. Select **DX00000004** from the **Starting Project Number** drop-down list.

Note: The **Ending Project Number** field assumes the value you entered in the Starting Project Number field.
3. Select the **Cost Codes** radio button to indicate which plan the project totals will be rolled from.
4. Clear the **Roll Plan Revenue Amounts** check box, if selected.
5. Click **Process**. A dialog box opens with the message, "1 Project(s) were processed."
6. Click **OK**.
7. Close the **Roll Up Project Cost** form.

Part 3: Verify WIP dollar values from the Projects form

1. Click the **Cost** tab on the **Projects** form.
2. View the **Project Costs**.
3. Write the values for each project cost here:

Project Cost	Value
Material	
Labor	

Other	
Overhead	
G & A	
Total Cost	
Total WIP Relieved	
Net WIP	

Part 4: Process the Print/Post Project WIP Relief activity

1. Select **Master Explorer > Modules > Project > Activities > Print/Post Project WIP Relief**. The **Print/Post Project WIP Relief** form opens.
2. Select **DX00000004** from the **Starting Project Number** drop-down list.

Note: The **Ending Project Number** assumes the values entered in the Starting Project Number field.

3. Leave the **WIP Start Date** and **Post Date** with the default values.
4. Click **Process**. A dialog box opens with the message, "1 Project(s) were processed."
5. Click **OK**.

Notice the values in the **Debit** and **Credit** columns in the grid.

6. Select the **Commit** radio button.
7. Click **Process**. A dialog box opens with the message, "1 Project(s) were processed."
8. Click **OK**.
9. Close the **Print/Post Project WIP Relief** form.

Part 5: Review WIP Relief

1. Return to the **Projects** form.
2. Select **Actions > Refresh current**.
3. Click the **Cost** tab.

Note: The Total WIP Relieved and Net WIP.

4. Click the **% Complete** tab.

Notice the breakdown by Project Cost Codes for both Actual Cost and Forecast Cost.

5. Click the **% Complete Details** tab.
6. Review the breakdown by Project **Cost Codes** for both **Actual Cost** and **Forecast Cost** by **Month**.

Project WIP Adjustments

The **Project WIP Adjustments** activity adjusts WIP at the task level.

To make WIP adjustments, you would need to enter any adjustments to the five adjustment accounts. Then, you would click **Post**. The **Print/Post Project WIP Relief** activity displays.

To complete, process the **Print/Post Project WIP Relief** activity.



Check your understanding



What are the three WIP Relief Methods?





How is WIP relieved if the WIP Relief Method is Revenue Recognition?





Lesson 11: Work breakdown structures (WBS) and projects

Estimated time

.25 hours

Learning objectives

In this lesson, you will:

- Describe the purpose and functionality of the Work Breakdown Structures form.

Topics

- Work Breakdown Structures

Work Breakdown Structures

The **Work Breakdown Structures** form allows you to define a workload so that reporting can be performed for many projects or tasks. You can create, maintain, and delete a WBS for project reporting purposes.

A WBS requires an alphanumeric WBS name and an optional description.

A WBS in the application is at a higher level than a WBS in Microsoft Projects. One example for setting up a WBS in the application would be a government contract which may consist of many projects.

The following table describes the tabs on the Work Breakdown Structures form:

Pricing category	Description
% Complete	The % Complete tab displays Actual , Forecast , Cost to Complete , % Complete , and Budget fields for all cost codes for a particular work breakdown structure. This tab also displays any Overhead and G & A costs included in the work breakdown structure as well as totals for each column.
WBS	The WBS tab shows a collapsible and expandable tree view of the selected WBS record. The tree displays project, task, resource, and work breakdown information. Project, task, and work breakdown structures can be expanded to show children. Resources have no children.

Status

Choose either Planned or Released from the Status field drop-down list.

If ...	Then ...
Planned is chosen	Transactions cannot be entered for project tasks assigned to the WBS.
Released is chosen	Transactions can be entered for project tasks assigned to the WBS.

Transactions that are controlled are:

- Project Resource
- Labor
- WIP Relief
- Invoice Milestone
- Revenue Milestone posting
- Job Material and Labor Transaction posting
- Purchase Order Receiving
- A/P Voucher posting



Useful project reports

Useful reports are the Indented WBS Cost Report and the WBS Percent Complete Report. Both reports are accessed by selecting **Master Explorer > Modules > Project > Reports**.





Exercise 11.1: Analyze project information

In this exercise, you will analyze project information for a customer, Ting Tang Bicycles.

Exercise steps

Part 1: Create the WBS

1. Select **Master Explorer > Modules > Project > Work Breakdown Structures**. The **Work Breakdown Structures** form opens.
2. Add a new record. The value of **Planned** appears in the **Status** field.
3. Type *Ting Tang* in the **WBS** field.
4. Type *Ting Tang Bicycle Projects* in the **Description** field.
5. Click the **Items** tab.
6. Select **Project** from the **Reference Type** drop-down list.
7. Select **DX00000004** from the **Project** drop-down list.
8. Add a new record.
9. Select **Project** from the **Reference Type** drop-down list.
10. Select **DX00000005** from the **Project** drop-down list.
11. Save the record.

Part 2: View the WBS

1. Click the **% Complete** tab.
2. View the costs associated with each **Cost Code** for the WBS you entered.
3. Click the **WBS View** tab.
4. Select the **Plus Sign (+)** beside **Detail**.
5. Expand the tree view by selecting the **Plus Sign** beside each **Reference** type.

6. View the resources associated with **Task 30**.

Note: Tasks accompany Projects and Resources accompany Tasks in WBS.

Part 3: Change the status of the WBS

1. Click the **Items** tab.
2. Select **Released** from the **Status** field drop-down list.
3. Save the record.
4. Close the **Work Breakdown Structures** form.

Check your understanding



What is the purpose of creating a WBS?





Where can you view the Actual, Forecast, Cost to Complete, % Complete, and Budget for all cost codes for a particular work breakdown structure?





Lesson 12: Revenue recognition and projects

Estimated time

.75 hours

Learning objectives

In this lesson, you will:

- Identify where in the application to set up revenue recognition for projects.
- Describe how to create revenue milestones for projects.
- State how to print/post project revenue milestones.

Topics

- Project Parameters – General tab
- Distribution Accounts maintenance
- Projects –Revenue tab
- Revenue Milestones
- Revenue Milestones – Revenue Recognition
- Revenue Milestones – Requirements tab
- Revenue Milestones – Audit
- Revenue Milestones Nomination
- Print/Post Project Invoice Milestones

Project Parameters – General tab

The **Project Parameters** form, **General** tab allows you to define default and project criteria for all projects.

The following table describes the fields and check box on the **Project Parameters** form, **General** tab that pertain to revenue recognition:

Field /check box	Description
Revenue Calculation	This field determines the method is used to calculate revenue when the milestone is nominated. The options are: Manual, % Over Milestone Cost, % Of Project Revenue, and Custom.
Cost Calculation	The Cost Calculation field determines the amount, and percent used for default values on the Projects form, Revenue tab.
Revenue Milestone Auditing	The Revenue Milestone Auditing check box, when selected, allows you to choose whether auditing should be turned on or off for the Revenue Milestones form.
Multi-Requirement Tasks	The Multi-Requirement Tasks parameter allows/does not allow tasks to be referenced by more than one revenue milestone at a time. The options are: • Never • Sometimes • Always
Unbilled A/R Account	This field contains the account number for unbilled Accounts Receivable (A/R) accounts. The unbilled A/R Account is debited during the posting of revenue milestones.



Distribution Accounts maintenance

A product code must be entered within the **Distribution Accounts** form for sales transactions related to the project. These accounts are credited when posting project revenue.

Projects – Revenue tab

The **Projects** form, Revenue tab allows you to set default milestone cost calculations and milestone revenue, and if applicable, percentages. These values are only used to set the default values on the **Revenue Milestones** form, **Revenue Recognition** tab.

Planned, **Actual**, and **Revenue** fields display as well as the **Calculated Margins** field. Planned fields can be manually set or automatically calculated by running the Project Cost Roll Up activity. Actual fields are updated when revenue milestones are posted.



Exercise 12.1: Set Revenue Recognition Milestone Calculation Methods

In this exercise, you will set Revenue Recognition Milestone Calculation Methods.

Exercise steps

Part 1: Verify WIP Relief method on your project

1. Return to the **Projects** form.
2. Select <your project number> from the grid view if not already selected.
3. Click the **WIP Relief** tab.
4. Verify **Revenue Recognition** is selected in the **WIP Relief Method** field.

Part 2: Select Revenue Milestone Calculation Methods

1. Click the **Revenue** tab.
2. Select **% of Project Cost** from the **Milestone Cost Calculation Method** drop-down list.
3. Type **10** in the **Percentage** field.
4. Select **% Over Milestone Cost** from the **Milestone Revenue Calculation Method** drop-down list.
5. Type **10** in the **Percentage** field.
6. Save the record.
7. Leave the form open as you will use it in the next exercise.

Revenue Milestones

The **Revenue Milestones** form is used to schedule revenues that are to be recognized for a project. You can specify values in the **Milestone Number** and **Milestone Description** fields, as well as notes related to the particular milestone. There are planned costs, actual costs and revenue for each milestone.

Posting a revenue milestone

Posting a revenue milestone is a two-step process:

1. Nominate the milestone
2. Post the milestone

If the milestone has not been posted, a **Calculate Cost/Revenue** button displays above the **Planned Cost** fields. By clicking this button, the **Cost** and **Revenue** fields calculate automatically based on the parameters set on the **Revenue Milestones** form, **Revenue Recognition** tab.



The WIP Relief Method must be set to Revenue Recognition on the **Projects** form, **WIP Relief** tab to access this form.



Revenue Milestones – Revenue Recognition

The **Revenue Milestones** form, **Revenue Recognition** tab defines the Milestone Cost Calculation and Milestone Revenue Calculation methods.

Milestone Cost Calculation

Defining the Milestone Cost Calculation method for a milestone indicates how project costs per milestone will be calculated.

The following table describes the options for Milestone Cost Calculation:

Option	Description
Manual	This option manually sets the cost to be used for WIP Relief during Revenue Recognition Milestone Nomination and Posting.
% of Project Cost	This option sets a percentage of the total planned cost from the Projects form, Revenue tab to be used during Revenue Recognition Milestone Nomination and Posting.
% of Milestone Revenue	This option sets a percentage of the total planned revenue from the Projects form, Revenue tab to be used during Revenue Recognition Milestone Nomination and Posting.
% of Milestone Requirements	This option sets a percentage of the total task costs from the Revenue Milestones form, Revenue Recognition tab to be used during Revenue Recognition Milestone Nomination and Posting.
Unrelieved Cost	This option calculates the cost that has accumulated since the last Revenue Recognition Milestone Nomination and Posting.
Custom	This is a reserved application code to be used if there are custom processing routines created.



Milestone Revenue Calculation

The Milestone Revenue Calculation method indicates how revenues are going to be calculated for the milestone.

The following table describes the options for Milestone Revenue Calculation:

Option	Description
Manual	This option manually sets the revenue to be used during Revenue Recognition Milestone Nomination and Posting.
% Over Milestone Cost	The percentage entered in the Percentage field serves as the markup of the milestone cost. This markup is the revenue that is recognized.
% of Project Revenue	This option multiplies the percentage entered in the Percentage field by the Total Planned Revenue from the Revenue tab of the Projects form.
Custom	This is a reserved code to be used if there are custom processing routines created.

Revenue Milestones – Requirements tab

The **Revenue Milestones** form, **Requirements** tab defines the required project tasks and resources for the milestone defined on the form. A description can be used to describe why this is a required task or resource. Using the **Notes** option on the **Main** toolbar may also be used to record more detailed information.

The **Required Complete** check box is used to indicate whether the requirement is a hard or soft requirement:

If...	Then...
The Required Complete check box is selected (hard requirement)	You would not be able to nominate the revenue recognition milestone until the requirement is met.
The Required Complete check box is not selected (soft requirement)	You would receive a warning when attempting to nominate the revenue recognition milestone that the requirement has not been met. However, you can continue with the nomination.

A task requirement necessitates that the task be marked with a status of “Complete.”

A resource requirement necessitates that the resource be completely issued to the project.



Revenue Milestones – Audit

When auditing, options are activated on the **Project Parameters** form, the user, sequence, date of change, and any of the values on the **Revenue Milestones** form, **General** or **Revenue Recognition** tabs are tracked.



Exercise 12.2: Create a revenue milestone

In this exercise, you will roll up your project costs and create a revenue milestone for your project.

Exercise steps

Part 1: Roll up costs on your project

1. Select **Master Explorer > Modules > Project > Utilities > Roll Up Project Cost**. The **Roll Up Project Cost** form opens.
2. Select **<your project number>** from the **Starting Project Number** drop-down list.
Note: The Ending Project Number field assumes the value you entered in the Starting Project Number field.
3. Select the check box for **Roll Plan Revenue Amounts**.
4. Click **Process**. A dialog box opens with the message, "1 Project(s) were processed."
5. Click **OK**.
6. Close the **Roll Up Project Cost** form. You are returned to the **Projects** form.

Part 2: Create a revenue milestone

1. Click **Revenue Milestones**. The **Revenue Milestones (Linked)** form opens.
2. Type **1** in the **Milestone Number** field.
3. Type **1st Revenue Milestone** in the **Description** field.
Note: This field is not labeled. It is located to the right of the Milestone Number field.
4. Type **<today's date>** in the **Planned Complete Date** field.
5. Click the **Revenue Recog** tab.
6. Verify **% of Project Cost** is selected in the **Method** field.
7. Verify **10** is in the **Percentage** field.
8. Verify **% Over Milestone Cost** is selected in the **Method** field.
9. Verify **10** is in the **Percentage** field.
10. Click the **General** tab.
11. Save the record.
12. Click **Calculate Cost/Revenue**. Note that values now display in the **Planned** fields.
13. Verify the **Cost** fields are calculated at 10% of the Planned Costs on the **Revenue** tab.

14. Verify the **Revenue** field is calculated at 10% over the Planned Costs on the **Revenue** tab.

15. Save the record.

Revenue Milestone Nomination

The **Revenue Milestone Nomination** form is used to nominate and un-nominate revenue recognition milestones, as well as change the cost/revenue that will post when processing the Print/Post Project Revenue Milestones activity.

The bottom half of the form automatically updates. As each milestone is highlighted, the milestone Period information updates, while the current Period information remains the same. This is due to the current accounting period not changing as you browse the milestones.

The two columns of numbers at the bottom of the form display the total and nominated totals for the milestone accounting period and the current accounting year. The top row (Period and Year) shows all milestones regardless of nomination status, and the bottom row (Nomination :) shows only those milestone totals which are nominated.

The Auto Nomination button allows you to have the system check all the milestone requirements to see if they have been fulfilled. If so, the milestone is automatically set to "Nominated." The Planned Complete Date compared to the Current Date is also a criteria used by the auto-nomination process.

You can nominate/un-nominate a milestone by selecting and clearing the Nominated check box.

There are two Calculate/Cost Revenue buttons, one for Planned costs and one for Actual costs. Clicking either one of these buttons will automatically re-calculate each date, cost and revenue fields based on the parameters set up on the Revenue Milestones form, Revenue Recognition tab.



Print/Post Project Revenue Milestones

The **Print/Post Project Revenue Milestones** activity allows you to recognize revenue for the project while also relieving WIP.

Example journal transactions

The tables below are examples of revenue milestones are affected for the two types of journal transactions.

Revenue milestone transaction – Project Control Ledger Distribution Journal

Below is an example of how a revenue milestone affects the Project Control Ledger Distribution Journal:

	Debit	Credit
COGS - Material	100.00	
Project Account - Material		100.00

Revenue milestone transaction – Accounts Receivable Distribution Journal

Below is an example of a revenue milestone posting transaction as it affects the Accounts Receivable Distribution Journal:

	Debit	Credit
Unbilled A/R Account	200.00	
Sales Revenue		200.00



Exercise 12.3: Nominate and post a revenue milestone

In this exercise, you will nominate and post a revenue milestone that you created in the previous exercise 12.2.

Exercise steps

Part 1: Nominate the project revenue milestone

1. Select **Master Explorer > Modules > Project > Revenue Milestone Nomination**. The **Revenue Milestone Nomination (Filter In Place)** form opens.
2. Select **<your project number>** from the **Project** drop-down list.

3. Run **Filter In Place**.
4. Review the fields under the first **Calculate Cost/Revenue** button (left side) of the form. These fields represent planned values. Values display in all the planned cost fields.
5. Click the **Calculate Cost/Revenue** button in the **Actual** section (right side) of the form. Values display in all the actual cost fields.
6. Save the record.
7. Click **Auto Nomination**. A dialog box opens with the message, "X Revenue Milestone(s) were selected."

Note: You may have to set your current year and period.

8. Click **OK**.
Notice the **Nominated** check box has been selected.
9. Close the **Revenue Milestone Nomination** form. You are returned to the **Revenue Milestones (Linked)** form.

Part 2: Post/Print the project revenue milestone

1. Click **Post Milestone**. The **Print/Post Revenue Milestones** form opens with your project, milestone number, period and year.
2. Click **Process**. A dialog box opens with the message, "1 Project(s) were processed."
3. Click **OK**.
4. Select the **Commit** radio button.
5. Click **Process**. A dialog box opens with the message, "1 Project(s) were processed."
6. Click **OK**.
7. Close the **Print/Post Project Revenue Milestones** form.
8. Close the **Revenue Milestones (Linked)** form. You are returned to the **Projects** form.
9. Select **Actions > Refresh current** to refresh your project information. Note: There is now data in the Actual fields.

Check your understanding



Where is Planned Revenue entered?





What needs to happen after revenue is entered on the Revenue Milestones form?





Lesson 13: Invoicing and projects

Estimated time

.75 hours

Learning objectives

In this lesson, you will:

- Discuss how to set up invoicing for projects.
- Describe how to create invoice milestones for projects.
- Explain how to post project invoices.

Topics

- Project Retention Codes
- Project Parameters – General tab
- Projects – Invoicing tab
- Projects – Tax Info tab
- Invoice Milestones
- Invoice Milestones- Requirements tab
- Invoice Milestones - Audit
- Invoice Milestones Nomination
- Post Project Invoice Milestones
- Invoices, Debit and Credit Memos
- Post Project Retention Invoice
- Invoices, Debit and Credit Memos G/L Distribution
- Invoice Posting

Project Retention Codes

Retention is most commonly used by government contractors for invoicing purposes. It allows the government agency being invoiced to retain some of the monies it owes until the project has reached a satisfactory conclusion.

Retention code example

A retention code may be set up which retains 15% until the project is 50% complete. The retention could then drop to 10% until the project is delivered. It could then drop to 2% for a final warranty period, which will be paid when the warranty period is over.

Project retention codes set up on the **Project Retention Codes** form are referenced on the Projects form, Invoicing tab. The retention code is alphanumeric and includes a Description and Billing Method.

The following table describes the fields and columns on the **Project Retention Codes** form:

Field	Description
Billing Method	The Billing Method field options are: • Individual – Use this option for customers who want a separate retention invoice for every invoice they receive. • Total – Use this option for customers that want retention held and not billed until the project is complete.
Percent Complete	The percent of the project to be completed is typed in this column. When the project reaches this level of completion, the retain percentage displays in the Retention Percent column.
Retention Percent	The percent of the invoice that will be held back as the retention is typed in this column.



Project Parameters – General tab

The **Project Parameters** form, **General** tab allows you to define default and project criteria for all projects.

The following table describes the fields and check boxes on the **Project Parameters** form, **General** tab that pertains to invoicing:

Field/check box	Description
Invoice Method	The Invoice Method selected from the drop-down list is the default value of the Invoice Method field on the Projects form, Invoicing tab.
Default Retention Code	This setting is the default Retention Code for all projects. The code automatically displays on the Projects form, Invoicing tab.
Invoice Milestone Auditing	This parameter's check box allows you to choose whether auditing should be turned on or off on the Revenue Milestones form.
Auto Invoice Milestone Nominate	This parameter's check box determines the default flag for the Auto Nominate Milestone check box on the Projects form, Invoicing tab.
Multi-Requirement Tasks	This parameter's drop down list allows or does not allow tasks to be referenced by more than one revenue milestone at a time. The options are: • Always • Sometimes • Never
Invoiced Retention Account	The general ledger account number value that displays in the Invoiced Retention Account field is used in journal entries when invoices are posted to the A/R Distribution Journal.
Unbilled A/R Account	The general ledger account number value that displays in the Unbilled A/R Account field is used in journal entries when invoices are posted to the A/R Distribution Journal.
Unbilled Retention Account	The general ledger account number value that displays in the Unbilled Retention Account field is used in journal entries when invoices are posted to the A/R Distribution Journal.
Advance Payment Account	The general ledger account number value that displays in the Advance Payment Account field is used in journal entries when invoices are posted to the A/R Distribution Journal.



The Invoiced Retention Account and Unbilled Retention Account are mostly used for government contracts where some of the monies are retained until the project has reached a satisfactory conclusion.

Projects – Invoicing tab

The **Projects** form, Invoicing tab displays invoicing, retention, accumulative totals, and exchange rate fields.

Invoicing tab, Invoicing section

The following table describes the check box and fields in the Invoicing section of the **Projects** form, **Invoicing** tab:

Field/check box	Description
Auto Nominate Invoice	The Invoice Milestone is automatically nominated when it is created when this check box is selected.
Invoice Method	The Invoice Method defaults from the Project Parameters form. The Invoice Method options can be changed. The options are as follows: • Manual – Allows you to maintain the Invoice Milestone manually. This is typically used when invoicing and revenue recognition are not tightly coupled. • Revenue Recognition – the application automatically creates an Invoice Milestone when a Revenue Recognition Milestone is posted for the project. The Invoice Milestone is created with an invoice amount equal to the actual revenue posted on the Revenue Recognition Milestone form.
Invoice Terms	Invoice Terms default from the customer master billing terms. These are used for the terms codes during posting of the invoice milestone.

Invoicing tab, Retention section

The following table describes the fields in the **Retention** section of the **Projects** form, **Invoicing** tab:

Field	Description
Retention Code	The code selected from the Retention Code field is used to calculate retention during the invoice posting process.
Retention Terms	The code selected from the Retention Terms field is the terms that are used on any retention invoices for the project.

Invoice tab, Exchange Rate section

If multi-currency is enabled in your application database, you will have the option to use a fixed exchange rate for the duration of the project.

Translate

If multi-currency is enabled in your application database, you have the option to use the **Translate icon**, which will convert all Project costs and revenues to your domestic currency.

These 6 forms have the Translate Icon:

- Revenue Milestones
- Revenue Milestone Nomination
- Invoice Milestones
- Invoice Milestone Nomination
- Estimate Revenue Milestones
- Estimate Invoice Milestones



Exchange rates reference

See **System > Help** or the CloudSuite Industrial: **Using Multi-Currency** training workbook for more information regarding exchange rates.



Invoicing tab, Accumulative section

The following table describes the fields in the **Accumulative** section of the **Projects** form, **Invoicing** tab:

Field	Description
Total Retention Project to Date	This field displays the total retention amount maintained for the current project.
Total Invoiced Project to Date	This field displays the total amount invoiced for the current project to date.

Projects – Tax Info tab

The **Projects** form, **Tax Info** tab displays pertinent tax codes customers using an area based tax system. An item based tax system would display tax information at the project task resource level.





Exercise 13.1: Add retention terms

In this exercise, you will add retention terms for your project.

Exercise steps

1. Return to the **Projects** form.
2. Click the **Invoicing** tab.
3. Select **N45 – Net 45 Days** from the **Retention Terms** drop-down list.
4. Save the record.

Invoice Milestones

The **Invoice Milestones** form is used to schedule invoices to send to the customer. You can specify a Milestone Number and Milestone Description, as well as notes related to the particular milestone. There are planned invoice amounts and actual invoice amounts for each milestone.

Posting an invoice milestone is a two-step process:

1. Nominate the milestone
2. Post the milestone

Invoice Milestones, General tab

The following table describes a field and check box on the **Invoice Milestones** form, **General** tab:

Field/check box	Description
Planned Complete Date	The Planned Complete Date is the date you expect to print and post the milestone.
Tax Code	The Tax Code field can be overridden at the time of creating milestones for both area and item taxing if the Prompt on Line Item field has been activated on the Tax Systems form.
Ignore Retention on Invoice	This check box is used to indicate that whether retention is to be calculated on the invoice. Select this field if the milestone is not subject to retention. Clear this field if the milestone is subject to retention. This selection defaults from the project Revenue Milestone form if it was automatically created due to the Invoice Method of Revenue Recognition on the Projects form, Invoicing tab.



The WIP Relief Method must be set to Revenue Recognition on the Projects form, WIP Relief tab in order to access this form.



Invoice Milestones, Requirements tab

The **Invoice Milestones** form, **Requirements** tab allows you to specify tasks and resources that are requirements for the milestone. A description should be used to explain why this is a required task or resource. Using the **Notes** option on the **Main** toolbar may also be used to record more detailed information.

The **Required Complete** check box is used to indicate whether the requirement is a hard or soft requirement:

If...	Then...
The Required Complete check box is selected (hard requirement)	You would not be able to nominate the invoice milestone until the requirement is met.
The Required Complete check box is not selected (soft requirement)	You would receive a warning when attempting to nominate the invoice milestone that the requirement has not been met. However, you can continue with the nomination.

A task requirement necessitates that the task be marked with a status of "Complete."

A resource requirement necessitates that the resource be completely issued to the project.



Invoice Milestones, Audit tab

The audit trail, when activated on the Project Parameters form, tracks any changes made to the amounts, user name, or dates for invoice milestones.



Invoice Milestone Nomination

The **Invoice Milestone Nomination** form is used to nominate and un-nominate invoice milestones, as well as change the invoice amount and retention flag that will post when processing the Invoice Milestone Posting activity.

The bottom half of the form automatically updates. As each milestone is highlighted, the milestone Period information updates, while the current Period information remains the same. This is due to the current accounting period not changing as you browse the milestones.

The two columns of numbers at the bottom of the form display the total and nominated totals for the milestone accounting period and the current accounting year. The top row (Period and Year) shows all milestones regardless of nomination status, and the bottom row (Nominated) shows only those milestone totals which are nominated.

You can nominate/un-nominate a milestone by selecting or clearing the **Nominated** check box.

You can override the Actual Date and/or Actual Invoice Total prior to posting the milestone.

Information in the **Taxable Amount** field defaults to the information in the **Actual Invoice Amount** field. This can be overridden at the time of nomination for tax purposes.



Post Project Invoice Milestones

The **Post Project Invoice Milestones** activity allows you to post the invoice milestones to the Accounts Receivable **Invoices, Debit and Credit Memos** form.

A report is printed showing amounts to be debited and credited. The Project Invoice and Retention Invoice can also be printed now.



Journal entries are not created when the Post Project Invoice Milestones prints. Journal entries are created when the Invoices, Debit, and Credit Memos are posted to the Accounts Receivable Distribution Journal.



Invoices, Debit and Credit Memos

An invoice or retention invoice can be viewed on the **Invoices, Debit and Credit Memos** form after generating either an invoice or retention invoice for the project.

When invoices are added as part of the posting process from Customer Order Entry, the **Posted From Order** check box will be selected. The project will be referenced in the **Order** field. The **Notes** field specifies the type of invoice.

Example of a project invoice with retention code billing method of individual

Progressive Industries - Dallas 517 S 2nd Ave Dallas TX 75226 214-741-6700		Invoice Number: 445 Page:1 Date:4/7/2017 Project Invoice Fed ID: 43-2765936 Currency: USD US Dollar				
Fed ID: 12						
B Mike Kehl I Lake House Cycles L 700 Front St L PO Box 4 T Binghamton NY 13905 O USA	S Peter Paulin H Lake House Cycles I 700 Front St P PO Box 4 T Binghamton NY 13905 O USA					
Fax: 607-723-1209 Proj Contract LHC-1020						
Order	Purchase Order	Packages	Prepaid	Weight	Ship Via	Terms
DX00000009						Net 45 Days
Line/Rel	Qty Ordered	Qty Shipped	Qty Back Ordered	Unit Price	Extended Price	
1	Milestone: 1					
	Description: First invoice					
	Amount:	5,216.38	Net: 4,694.74			
	Retention:	521.64				
Advance Payment Summary:						
Advance Payment Amount: 0.00						
Less Accumulated Deduction: 0.00						
Remaining Advance Payment: 0.00						
Sales Amount 4,694.74						
Freight 0.00						
Misc Charges 0.00						
Sales Tax 0.00						
Prepaid Amount 0.00						
Total 4,694.74						
Gracias por hacer negocios con los ciclos Progressive Unidos. Nuestros nuevos ciclos de la serie XT Racing ya están disponibles! Vea el folleto adjunto para más detalles.						

Project invoice with retention code billing method of individual

Example of a retention invoice

Progressive Industries - Dallas 517 S 2nd Ave Dallas TX 75226 214-741-6700		Invoice Number: 446 Page: 1 Date: 4/7/2017 Retention Invoice				
Fed ID: 12		Fed ID: 43-2765936 Currency: USD US Dollar				
B Mike Kehl I Lake House Cycles L 700 Front St P PO Box 4 T Binghamton NY 13905 O USA	S Peter Paulis H Lake House Cycles I 700 Front St P PO Box 4 T Binghamton NY 13905 O USA					
Fax: 607-723-1209 Proj Contract LHC-1020						
Order	Purchase Order	Packages	Prepaid	Weight	Ship Via	Terms
DX00000009						Net 45 Days
Line/Ref	Qty Ordered	Qty Shipped	Qty Back Ordered	Unit Price	Extended Price	
1						
Milestone: 1 Description: First invoice						
Net: 521.64						
Advance Payment Summary: Advance Payment Amount: 0.00 Less Accumulated Deduction: 0.00 Remaining Advance Payment: 0.00						
Sales Amount						521.64
Freight						0.00
Misc Charges						0.00
Sales Tax						0.00
Prepaid Amount						0.00
Total						521.64

Gracias por hacer negocios con los ciclos Progressivos
 Usted. Nuestros nuevos ciclos de la serie XT Racing ya están disponibles! Vea el folleto adjunto para más detalles.

Retention invoice

Example of a project invoice with retention code billing method of total

Progressive Industries - Dallas 517 S 2nd Ave Dallas TX 75226 214-741-6700		Invoice Number: 448 Page:1 Date:4/7/2017 Project Invoice	
Fed ID: 12		Fed ID: 43-2765936 Currency: USD US Dollar	
B Mike Kohl I Lake House Cycles L 700 Front St L PO Box 4 T Binghamton NY 13905 O USA	S Peter Pauls H Lake House Cycles I 700 Front St P PO Box 4 T Binghamton NY 13905 O USA		
Fax: 607-723-1209 Proj Contract LHC-1020			

Order	Purchase Order	Packages	Prepaid	Weight	Ship Via	Terms
DX00000006						Net 45 Days

Line/Rel	Qty Ordered	Qty Shipped	Qty Back Ordered	Unit Price	Extended Price
1					

Milestone: 1
Description: Final Invoice less retention

Net: 112,500.00

Project Invoice Summary:

Milestones On Current Invoice:	125,000.00
Milestones Previously Invoiced:	0.00
Project To Date M/S Invoiced:	125,000.00
Less Retention:	12,500.00
Total Eligible To Invoice:	112,500.00
Less Previously Invoiced:	0.00
Less Advance Payment:	0.00
Net Invoice Amount:	112,500.00

Advance Payment Summary:

Advance Payment Amount:	0.00
Less Accumulated Deduction:	0.00
Remaining Advance Payment:	0.00

Sales Amount	112,500.00
Freight	0.00
Misc Charges	0.00
Sales Tax	0.00
Prepaid Amount	0.00
Total	112,500.00

Gracias por hacer negocios con los ciclos Progressivos
 Usted. Nuestros nuevos ciclos de la serie XT Racing ya están
 disponibles! Vea el folleto adjunto para más detalles.

Project invoice with retention code billing method of total

Invoices, Debit and Credit Memos G/L Distribution

Invoice distribution records can be created and viewed before invoices and accounts are posted to the Accounts Receivable Distribution Journal using the **Invoices, Debit and Credit Memos G/L Distribution** form. The Unbilled A/R Account will be credited for normal project invoices.



Post Project Retention Invoice

The Post Project Retention Invoice is used to create the final invoice for any outstanding retention on the project. The invoice created is called a Liquidation Invoice.

Access the form by selecting **Master Explorer > Modules > Project > Post Project Retention Invoice**.



Journal entries are not created when Retention Invoices print. Journal entries are created when A/R Invoices, Debit, and Credit Memos are posted to the Accounts Receivable Distribution Journal.



Invoice Posting

The **Invoice Posting** activity is used to create and view journal entries in the Accounts Receivable Distribution Journal. Project invoice posting does not create entries in the Project Control Distribution Journal. For revenue recognition for a project, manual adjustments entries must be made for accounts receivable.

Example journal transactions

Examples of journal transactions as they affect the Accounts Receivable Distribution Journal are displayed in the following tables.

Project invoice posting transaction – Accounts Receivable Distribution Journal

Below is an example of a project invoice posting transaction as it affects the Accounts Receivable Distribution Journal:

	Debit	Credit
Accounts Receivable	1000.00	
Unbilled A/R		1000.00

Retention invoice posting transaction – Accounts Receivable Distribution Journal

Below is an example of a retention invoice posting transaction as it affects the Accounts Receivable Distribution Journal:

	Debit	Credit
Invoiced Retention Account	1000.00	
Unbilled Retention Account		1000.00

Liquidation invoice posting transaction – Accounts Receivable Distribution Journal

Below is an example liquidation invoice posting transaction as it affects the Accounts Receivable Distribution Journal:

	Debit	Credit
Invoiced Retention Account	1000.00	

	Debit	Credit
Unbilled Retention Account		1000.00



Exercise 13.2: Post an invoice milestone

In this exercise, you will post the invoice generated from the Revenue Milestone created in Exercise 12.2. You will then find the invoice in the Invoices, Debit and Credit Memos form.

Exercise steps

Part 1: View the invoice milestone created from Revenue Recognition

1. Return to the **Projects** form.
2. Click the **Invoicing** tab.
3. Note the Auto Nominate Invoice check box is selected.
4. Click **Invoice Milestones**. The **Invoice Milestones (Linked)** form opens.
5. Note the invoice was automatically created when posting revenue milestone number 1. This is because the Invoice Method is Revenue Recognition.
6. Note that the invoice is nominated but not posted.
7. Close the **Invoice Milestones (Linked)** form.
8. Close the **Projects** form.

Part 2: Post the invoice milestone

1. Select **Master Explorer > Modules > Project > Activities > Post Project Invoice Milestones**. The **Post Project Invoice Milestones** form opens.
2. Verify the *<current period>* is in the **Ending Milestone Period** field.
3. Select *<your project number>* from the **Starting** field **Project** drop-down list.
Note: The **Ending** field assumes the value you entered in the **Starting** field.
4. Type *1* for both **Starting** and **Ending** values in the **Invoice Milestone** fields.
5. Select the **Post** check box.
6. Verify the **Project** radio button is selected.
7. Click **Process**.
8. Close the **Report Viewer**. A dialog box opens with the message, "Report Submitted."
9. Click **OK**.
10. Close the **Post Project Invoice Milestones** form.

Part 3: View the invoices in the Invoices, Debit and Credit Memo form

1. Select **Master Explorer > Modules > Finance > Accounts Receivable > Invoices, Debit and Credit Memos**. The **Invoices, Debit and Credit Memos (Filter In Place)** form opens.
2. Select **12-Lake House Cycles** from the **Customer** drop-down list.
3. Run **Filter In Place**.
4. Note the following on line **2**:
 - **Project Number** is referenced in the **Order** field.
 - **Posted from Order** check box is selected.
 - **A/R Account** is 11000 – Accounts Receivable.
5. Select line **3** from the grid.
6. Note the following on line **3**:
 - **A/R Account** is 11030-Invoiced Retention.
7. Close the **Invoices, Debit and Credit Memos** form.

Part 4: View the invoice

1. Select **Master Explorer > Modules > System > Report Output Files**. The **Report Output Files (Filter In Place)** form opens.
2. Run **Filter In Place**.
3. Select the **PostProjectInvoiceMilestonesLaser** report.
4. Click **View**. The invoice displays.

Note: The first invoice is the Retention invoice and the second invoice will be the Project invoice.
5. Review each of the invoices.
6. Close the **Report Viewer**.
7. Close the **Report Output Files** form.

Check your understanding



What are the two Invoice Methods?





True or False. Posting project invoices from the A/R Invoices, Debits and Credits form will create journal entries in both the Accounts Receivable Distribution Journal and the Project Control Distribution Journal.





Lesson 14: Payroll and projects

Estimated time

.25 hours

Learning objectives

After completing this lesson, you will be able to:

- Describe how to generate payroll with labor transactions associated to a project.

Topics

- Generate Payroll

Generate Payroll

Labor hour transactions, which have been posted directly to a project, can be summarized when generating payroll labor transactions by selecting the **Summarize Project Transactions** check box on the **Generate Payroll** form.

Access the **Generate Payroll** form by selecting **Master Explorer > Roles > Payroll > Activities > Generate Payroll**.





Lesson 15: Project Manager Home

Estimated time

.25 hours

Learning objectives

In this lesson, you will:

- Describe Project Manager Home.
- Describe the information that can be accessed via the Navigator tab within Project Manager Home.

Topics

- Project Manager Home - Dashboard
- Project Manager Home- Navigator
- Project Manager Home – Tasks
- Project Manager Home – Flow

Project Manager Home – Dashboard

The Project Manager Home form provides a wide variety of information from other parts of the application in one form, all intended to be used by a project manager.

Access the **Project Manager Home** form by selecting **Master Explorer > Modules > Project > Project Manager Home**.



The **Project Manager Home** form is not part of a permission group. Your system administrator must give you authorization to get access to this form.

Icons/buttons

The icons/buttons that appear on the top of the form can be used to quickly open complimentary applications. If an application is not installed, the icon/button is disabled.

The following table describes the icons/buttons on the **Project Manager Home** form, **Dashboard** tab:

Icons/buttons	Description
Project	Opens Microsoft® Project
Word	Opens Microsoft® Word
Excel	Opens Microsoft® Excel
Outlook	Opens Microsoft® Outlook
Bing	Opens the default browser and navigates to the Bing® web page.
Infor 365 Support	Opens the default browser and navigates to the Infor365 Online Support web page.

The **Dashboard** tab on the **Project Manager Home** form provides links to forms and activities useful for a project manager.

The **My Stats** section of the form provides three links:

- Alerts
- Tasks
- Inbox

You are redirected to the **Tasks** tab when you select any one of these three links.

You can hover your mouse over any of the columns in the charts and graphs to see data values for that column.

Tabs

The Project Manager Home form has the following tabs:

- Navigator
- Tasks
- Flow



Project Manager Home – Navigator

The table below describes the field and tabs found on the **Navigator** tab on the **Project Manager Home** form. Most the information is read-only.

Field/tab	Description						
Project Manager	The Project Manager field defaults to the user ID logged into the application. The ID can be changed to another project manager ID or the field can be cleared to return all data. Note: This field is not used for the tabs listed in this table, however it is important to review the information to see how this field affects the data on each tab.						
Hot List	The Hot List tab displays project resource items that are potentially late. This tab contains three grids with the following information: • A list of late items • Project resources that match the selected item • Planning detail for the selected item The information on the tab defaults from the Planning Detail form. • If a Project Manager is selected, only project resource items assigned to the manager's projects display, and only the manager's project resources are shown. • The Cutoff Date field defaults to the current date and is not required. The field filters the Items grid. All items are cleared if you clear this field.						
Projects	The Projects tab displays the Projects, Tasks, and Resource grids with the following information: • A list of projects • Project tasks for the selected project • Project resources for the selected project The information comes from the Projects, Project Tasks, and Project Resources forms. • If a Project Manager is selected, only projects assigned to that project manager display in the Projects grid. Use the radio buttons to filter for data. The options are: <table border="1"> <thead> <tr> <th>Radio button</th><th>Description</th></tr> </thead> <tbody> <tr> <td>Past Due</td><td>Displays active projects where the current date is greater than the end date</td></tr> <tr> <td>Resources Late</td><td>Displays active projects where at least one task or resource has a "Late Alert"</td></tr> </tbody> </table>	Radio button	Description	Past Due	Displays active projects where the current date is greater than the end date	Resources Late	Displays active projects where at least one task or resource has a "Late Alert"
Radio button	Description						
Past Due	Displays active projects where the current date is greater than the end date						
Resources Late	Displays active projects where at least one task or resource has a "Late Alert"						

Field/tab	Description								
	<table> <tr> <td>Active</td><td>Displays all projects with a status of "Active"</td></tr> <tr> <td>Inactive</td><td>Displays all projects with a status of "Inactive"</td></tr> <tr> <td>Complete</td><td>Displays all projects with a status of "Complete"</td></tr> <tr> <td>All</td><td>Displays all projects</td></tr> </table>	Active	Displays all projects with a status of "Active"	Inactive	Displays all projects with a status of "Inactive"	Complete	Displays all projects with a status of "Complete"	All	Displays all projects
Active	Displays all projects with a status of "Active"								
Inactive	Displays all projects with a status of "Inactive"								
Complete	Displays all projects with a status of "Complete"								
All	Displays all projects								
WBS	The WBS tab displays the following WBS, WBS Item, and WBS % Complete information on three separate grids: • A list of WBSs • WBS Items for the selected WBS • WBS % Complete of the selected WBS The Project Manager field does not apply to this tab. The information that displays comes from the Work Breakdown Structures form.								
Jobs	The Jobs tab displays jobs that are referenced to projects on three separate grids with the following information: • A list of jobs • Job operations for the selected job • Materials for the selected operation The information comes from the Job Orders, Job Operations, and Job Materials forms. • If a project manager is selected, only jobs assigned to the project manager display. • The Cutoff Date field defaults to the current date and is not required. • The field is used to filter the Jobs grid. All jobs are shown by clearing this field. • Select values in the WBS drop-down list to filter the Jobs grid by WBS.								
Purchase Orders	The Purchase Orders tab displays purchase order information. A list of purchase order lines associated to projects display. The information comes from the Purchase Orders form. • If a project manager is selected, only purchase orders that reference projects assigned to the project manager display. • The Cutoff Date field defaults to the current date and is not required. • The field is used to filter the Purchase Orders grid. • All purchase order lines tied to a project display by clearing this field. • Select values in the WBS drop-down list to further filter by WBS.								

Field/tab	Description
Shipping	The Shipping tab displays a list of project resource shipments. The information comes from the Project Resource Shipping form and displays all shipments where the number in the Qty Shipped field is less than the Qty Required field. • If a project manager is selected, only shipments that reference projects assigned to the project manager display. • The Cutoff Date field defaults to the current date and is not required. If a cutoff date is selected, only shipments which are late or projected to be late on or before the cutoff date are listed. • All shipments display if the cutoff date field is cleared. • If a WBS is selected, only shipments that reference projects that are assigned to that WBS display.
Transactions	The Transactions tab displays project transactions. The information comes from the Project Transactions form. • If a Project Manager is selected, only transactions for projects assigned to the project manager display. • Select the Transaction Types check boxes to filter for data. Transaction Types are: ○ Accounts Payable ○ Job Close ○ Journal/Adjustment ○ Labor ○ Material ○ All Types Note: The All Types check box is selected by default.
Exceptions	The Exceptions tab displays planning exceptions. Select a project resource item from the Project Resources grid to view the exceptions for the selected item in the Exceptions grid. The information comes from the Planning Detail form. • If a project manager is selected, only project resources assigned to the project manager display.

Field/tab	Description
Cost Code Analysis	The Cost Code Analysis tab displays cost code information for projects on three separate grids including: • A list of projects • Cost code summary information for the selected project • Cost code detail information for the selected cost code The information comes from the % Complete and % Complete Details tabs on the Projects form. • If a project manager is selected, only projects assigned to the project manager display. • Select either the Project Cost Code radio button or the Job Cost Codes radio button to show the project cost codes or job cost codes associated to the project.
Costing Activity	The Costing Activity tab displays cost code amounts for 12 periods. The information that displays comes from the Project Transactions form. Select values from the Period Type field to specify if the data displays by: • Day • Month • Week • Year The Period Type field is a required field.
Period Costs	The Period Costs tab displays a graph for the cost of the selected project. The information comes from the Project/Job Period Costs form. If a project manager is selected, only projects assigned to the project manager display. The following fields and options are selected to filter the data on the graph: • Project Cost Codes radio button or Job Cost Codes radio button • Starting Year field that is used to begin counting the months • Starting Month (of the Starting Year) field from which to begin counting the months • Number of Months field that are graphed from the Starting Year/Starting Month After changing the selection criteria, click the Refresh button to view a new graph.
Accumulated Costs	The Accumulated Costs tab displays a graph for the cost of the selected project. The information comes from the Project/Job Accumulated Cost by Period form. The Project Manager field does not apply to this tab. The following fields and options are selected to filter the data on the graph: • Project Cost Codes radio button or Job Cost Codes radio button • Starting Year field that is used to begin counting the months

Field/tab	Description
	• Starting Month field (of the Starting Year) from which to begin counting the months • Number of Months field that are graphed from the Starting Year/Starting Month After changing the selection criteria, click Refresh to view a new graph.



Project Manager Home – Tasks

The table below describes the three sections found on the **Tasks** tab on the **Project Manager Home** form.

Section	Description								
Alerts	The fields in the Alerts section are: • Past Due Projects • Past Due Project Tasks • Estimates to Expire Numbers display in either red or green colored font. The number is red if it above zero. The number is green if it is zero. <table><tr><th>If...</th><th>Then....</th></tr><tr><td>Past Due Projects is selected and the number is red</td><td>The Projects form opens, filtered on past due projects.</td></tr><tr><td>Past Due Project Tasks is selected and the number is red"</td><td>The Project Tasks form opens, filtered on past due project tasks.</td></tr><tr><td>Estimates to Expire is selected</td><td>The Estimate Projects form opens.</td></tr></table>	If...	Then....	Past Due Projects is selected and the number is red	The Projects form opens, filtered on past due projects.	Past Due Project Tasks is selected and the number is red"	The Project Tasks form opens, filtered on past due project tasks.	Estimates to Expire is selected	The Estimate Projects form opens.
If...	Then....								
Past Due Projects is selected and the number is red	The Projects form opens, filtered on past due projects.								
Past Due Project Tasks is selected and the number is red"	The Project Tasks form opens, filtered on past due project tasks.								
Estimates to Expire is selected	The Estimate Projects form opens.								
Tasks	The Tasks section lists the tasks assigned to the logged in user.								
Inbox	The Inbox section lists the messages for the logged-in user. • Double-click a field in the Inbox grid to view all the emails that share that value. • All Inbox fields can be filtered except for those marked as Read and Expired.								



More information about tasks

See **Adding and Viewing User Tasks** in **System > Help** for more information about tasks.
Access the Inbox form for more information about the Inbox.

Project Manager Home – Flow

The **Project Manager Home** form, Flow tab contains a business process flow including links to forms and Help topics appropriate for each form.

- Double-click labeled items to open Help topics.
- Double-click icon squares to open forms.





Lesson 16: Microsoft Project

Estimated time

.5 hours

Learning objectives

In this lesson, you will:

- Recognize import/export capabilities between Infor CloudSuite and Microsoft Project.

Topics

- About Microsoft Project (MS Project)
- Project Tasks and MS Project

About Microsoft Project (MS Project)

Microsoft Project (MS Project) interfaces with CloudSuite Industrial.



Infor CloudSuite Industrial Microsoft® Office Integration User Guide

Chapter 4: Microsoft® Project Add-In for Infor CloudSuite

User must log in to Infor Xtreme, and locate the latest KB. This KB provides instructions on how to download the Infor CloudSuite Office Add-ins and install on the client machine.

You will need to follow the instructions within the Infor CloudSuite Industrial Microsoft Office Integration User Guide.

Sync In, Sync Out, and Enable Auto Sync Out

Sync In, **Sync Out**, and **Enable Auto Sync Out** are buttons found on the **Microsoft Project toolbar ADD-INS** tab.



ADD-INS and options within Microsoft Project

The items on the toolbar are enabled or disabled based on the following conditions.

- **Sync Out/Sync In** are enabled only when the “Project In Infor CloudSuite” check box is selected.
- **Sync In** is disabled when a valid project number is not available.
- **Assign Infor CloudSuite Resource** and **Assign Infor CloudSuite Work Resource** are enabled only when the Project In Infor CloudSuite check box is selected and a valid Project Number is assigned.

Sync In and Sync Out synchronize changes on the Project between Infor CloudSuite and the Microsoft Project client. Enable Auto Sync Out enables you to Sync Out changes automatically without having to click **Sync Out**.

Button	Description
Sync In	This button, when clicked, brings the changes from the CloudSuite Project.
Sync Out	This button, when clicked, sends any changes made in the current project to CloudSuite Project.
Enable Auto Sync Out	This button is a toggle button; it is either enabled or disabled. If enabled, every change made in Microsoft Project is updated in CloudSuite immediately when you save the Project.

Sync In Activities

A Sync In Process:

- Brings in new tasks, material resources, and work resources created in the project.
- Updates existing tasks, material resources, and work resources
- Sets predecessors

Infor CloudSuite Project Information

The Infor CloudSuite Project Information screen collects all mandatory data required to create a Project in Infor CloudSuite. This screen can be accessed in the following ways:

- From the Add-Inn menu bar, select **Infor CloudSuite > Infor CloudSuite Project Information**. You can access this screen in this manner anytime to update project information.
- When you click the Sync Out button from the ribbon. This screen appears only if any required data is missing.

This is an image of the screen.

The screenshot shows a Windows-style dialog box titled "Infor CloudSuite Project Information". At the top left, there is a checked checkbox labeled "Project In Infor CloudSuite". Below this, the form is organized into two main sections. The first section contains three rows of data: "Project" with a text box containing "PJ00000006" and a text box containing "Bicycle Racing Track"; "Customer" with a dropdown menu showing "12" and a text box containing "Lake House Cycles"; and "Ship To" with a dropdown menu showing "0" and a text box containing "Lake House Cycles". The second section contains three rows of data: "Current Date" with a date picker showing "7/ 3/2015"; "Start Date" with a date picker showing "7/ 3/2015"; and "End Date" with a date picker showing "8/26/2015". To the right of these are three more rows: "Type" with a dropdown menu showing "FIX"; "Status" with a dropdown menu showing "Active"; and "Product Code" with a dropdown menu showing "PROJ-FG". At the bottom right of the dialog box, there are two buttons: "Sync Out" and "OK".

Infor CloudSuite Project Information

Project Tasks – MS Project

The **Project Tasks** form, **MS Project** tab allows you to set values needed to make the MS Project interface function work correctly.

MS Project tab

The following table describes the fields, check boxes and buttons on the **Project Tasks** form, **MS Project** tab:

Field/check box/button	Description
Predecessors	This field displays a list of items that must be completed prior to beginning the task.
Start Date	This field displays the start date for the Predecessor. This date defaults from the General tab.
End Date	This field displays the end date for the Predecessor. This date defaults from the General tab.
Firm Start Date	This check box must be selected to indicate that the Start Date is “frozen” and should not be changed.
Firm End Date	This check box, if activated, indicates the end date is “frozen” and should not be changed.
MS Project Export	This button must be clicked to create a Gantt chart in MS Project. *See note below.
MS Project Import	This button, when clicked activates the system to search for the appropriate MS Project session and reads the appropriate data back into the database.  The MS Project Export button and MS Project Import button do not display unless Microsoft Project is installed on the computer you are using.

MS Work Resources tab

The following table describes the fields on the **Project Tasks** form, **MS Work Resources** tab:

Field	Description
Resource Sequence	This field displays a list of items that must be completed prior to beginning the task.
Resource Type	This field displays a list of resource types. • Employee • User • Vendor
Resource Name	This field displays a list based on what is selected in the Resource Type field.





Demo: Microsoft Project Integration with Infor CloudSuite

This demonstration shows how to export the project data for into Microsoft Project, add a predecessor and import back into the application.

Demo steps

Part 1: Log in and add data to be recognized in Microsoft Project

1. Log on to the application on the computer that has Microsoft Project loaded and the necessary Infor CloudSuite Connection has been completed through **Infor CloudSuite > Setup** within Microsoft Project.
2. **Note:** For the demo, Instructors will log into to the CSI Click Once Client and will sign into CSI_Demo_DALS database.

Part 2: Go to the Projects form

1. Select **Master Explorer > Modules > Project > Projects**. The **Projects (Filter In Place)** form opens.

2. Run **Filter In Place**.
3. Select **DX00000006** from the grid on the left.

Part 3: Go to the Project Tasks form and add information specifically for MS Project

1. Click **Tasks** on the **Projects** form. The **Project Tasks (Linked)** form opens.
2. Confirm that **five project tasks** display in the grid.
3. Select task **10-Survey the site** from the grid.
4. Click the **MS Work Resources** tab.
5. Click the **Resource Type** drop-down list from the first row.
6. Select **Employee** from the **Resource** drop-down list.
7. Select **Sue-5** from the **Resource Name** drop-down list.
8. Select task **20-Excavate the site** from the grid.
9. Add the following employees as resources using the following table:

Resource Type	Resource Name
Employee	Brenda-8
Employee	Jerry-15

10. Save the record.

Part 4: Export data from the application to MS Project

1. Click the **MS Project** tab.
2. Click **MS Project Export**. A dialog box opens with the message, "[MS Project Export] was successful."
3. Click **OK**. Microsoft Project opens with your project information displaying in Microsoft Project.



The MS Project Export button and MS Project Import button do not display unless Microsoft Project is installed on the computer you are using.

Part 5: Review the project information and update a project task in Microsoft Project

1. Click the **Add-ins** tab.
2. Select **Infor CloudSuite > Infor CloudSuite Project Information** from the **Add-Ins menu bar** tab.
3. Review the information.
4. Click **OK**.
5. Click the **Predecessors** cell for the second task – **Excavate Track Site**.

6. Type **1** in the **Predecessors** cell. This is indicating that the task in row one should be completed before the second task.
7. Click the **Predecessors** cell for the third task – **Track Construction**.
8. Type **2** in the **Predecessors** cell. This is indicating that the task in row two should be completed before this task.
9. Click **Sync Out**. A dialog box opens with the message, "Sync Out process may modify or create new item(s) in Project, Project Task and Work Resources. Do you want to continue?"
10. Click **Yes**. A dialog box opens with the message, "Sync Out process completed successfully!"
11. Click **OK**.
12. Click **File > Save As** to save the project.
13. Type *Demo<the current date>.mpp* in the **File name** field
14. Save the record.

Part 6: Return to the application

1. Return to the **Project Tasks (Linked)** form.
2. Click **MS Project Import**. A dialog box opens with the message, "[MS Project Import] was successful."
3. Click **OK**.
4. Save the record.
5. Click **Excavate Track Site** from the grid.
Notice the value of **10** in the **Predecessors** field.
6. Click the **Track Construction** from the grid.
Notice the value of **20** in the **Predecessors** field.
7. Save the record.
Note: For the demo, this step is optional and needed only if you are going to send to someone else so they can view the project on their desktop using a viewer.
8. Close all forms.



Lesson 17: Project Completion Exercise

Estimated time

1 hour

Learning objectives

In this lesson, you will:

- Work with projects.

Topics

Project completion exercise



Scenario: Complete bicycle racing track project

It is now time to complete the bicycle racing track project you created in earlier exercises.

You are encouraged to experiment in this scenario to better understand system behavior.

For this scenario:

Project tasks 30 through 50 are almost complete.

- You will need to capture all costs associated with the three remaining project tasks. A large amount of labor cost must be posted to the tasks.
- Use the values in the table below to post labor costs:

Employee name	Task	Cost code	Labor hours
1 - David Daniels	30 -Construction	CL	120
9 – Jarrod A. Pau	30 - Construction	CL	120
11 - Jeff Smith	30 - Construction	CL	120
3 – Davender Konda	30 - Construction	CL	120
7 - Thomas Zimmerman	30 - Construction	CL	120
4 - Kirk Taylor	30 - Construction	CL	120
10 - Robert Penrod	40 - Landscaping	GL	40
8 - Brenda King	40 - Landscaping	GL	40
7 - Thomas Zimmerman	50 - Track Testing	GL	40
3 – Davender Konda	50 - Track Testing	GL	40

- View the Project Tasks form, % Complete tab for Actual Cost accumulation. If more detail is needed, print the Project Transactions form.
- Ship the remaining resources for the project.
- Create a revenue milestone, for project completion to relieve all unrelieved WIP costs and collect revenue of \$95,599.93 (an invoice will be created automatically). Ignore retention.
- Nominate and post the revenue milestone and post the invoice milestone.
- Change the status of your Job Order to “Complete”.

- Change the status of the project to “Complete.” No further processing will take place against this project in the future.
- Evaluate: Was the project profitable?



Course summary

Estimated time

.25 hours

Learning objectives

Now that you have completed this course, you should be able to:

- Describe Projects.
- Describe the purpose of cost codes.
- Describe estimate projects, estimate project tasks, and estimate project resources.
- Explain the purpose of the copy projects activity.
- Identify characteristics of projects, project tasks, and project resources.
- List options to assist with planning for project resources.
- Define characteristics of project resource transactions.
- Describe project labor transactions.
- Describe the benefit of cross-referencing purchase orders to a project.
- Explain how to create purchase orders cross-referenced to a project.
- Describe aspects of jobs orders cross-referenced to a project.
- Describe the purpose of using vouchers/adjustments for projects.
- List details required when entering a voucher adjustment distribution record.
- Discuss how to ship product directly from a project.
- Identify ways to create advance ship notices for a project.
- List the three WIP Relief Method options.
- Identify product code requirements for WIP Distribution Accounts.
- Explain how to alter the WIP Relief Method for individual project tasks.
- State how to print/post project WIP relief.
- Describe the purpose and functionality of the Work Breakdown Structures form.
- Identify where in the application to set up revenue recognition for projects.
- Describe how to create revenue milestones for projects.
- State how to print/post project revenue milestones.
- Discuss how to set up invoicing for projects.
- Describe how to create invoice milestones for projects.
- Explain how to post project invoices.
- Describe how to generate payroll with labor transactions associated to a project.
- Describe Project Manager Home.
- Describe the information that can be accessed via the Navigator tab within Project Manager Home.
- Recognize import/export capabilities between Infor CloudSuite and Microsoft Project.

Topics

- Course review

Check your understanding



Which of the following statements describes CloudSuite Project? Select all that apply.

- a) Project must be used with Microsoft Project.
- b) Project compares the actual cost accumulated on a project to the forecast cost of the project.
- c) Project is a cost management tool.
- d) The structure of Project makes it possible to assign planned costs and commitments by project, task, and resource providing the ability to manage project activities and costs.



What would be the pricing category to select on the Project Types form for a project with an agreed upon specific contract revenue amount?

- a) Calculated
- b) Fixed Price
- c) Non-billable
- d) Revenue



Which of the following statements are true about estimate project tasks? Select all that apply.

- a) Estimate project tasks can only be created manually.
- b) The Status field for an estimate project task is always "Estimate."
- c) Estimate project task numbers are incremented in steps of 10.
- d) Estimate project tasks can be referenced to Inventory, Purchase Order Requisitions, Purchase Orders, and Job Orders.



What tab on the Estimate Project Resources form is used to enter and view costs associated with each resource?

- a) Cost
- b) Default
- c) General
- d) % Complete



Which of the following statements are true about the Copy Projects activity? Select all that apply.

- a) The Copy Projects activity copies only estimate projects.
- b) The Copy Projects activity copies only the estimate project, estimate project tasks, and estimate project resources.
- c) The Copy Projects activity copies an estimate project, a project, or history project to a new or existing one.
- d) The Copy Projects activity copies the estimate project, estimate project tasks, estimate project resources, revenue milestones, and invoice milestones.



The information on the Project Tasks form, % Complete tab displays the amount of cost to complete the project, not the percent of the project completed.

- a) True
- b) False



What drop-down list on the Project Resources form, General tab references the origin of the item, such as Inventory, Purchase order Requisition, Purchase Order, or Job Order?

- a) Item
- b) Source
- c) Type
- d) Warehouse



Where does the application place item replenishment data for projects to review, edit, and firm into purchase orders, purchase requisitions, jobs, production schedules, Master Production Schedule (MPS) orders, and transfer orders?

- a) Material Planner Workbench Generation form
- b) Material Planner Workbench
- c) Planning Detail form
- d) Project Resources form



Which of the following statements are true about posting project resource transactions? Select all that apply.

- a) The cost and quantity of resources for a specific project can be issued to the project using the Project Resource Transactions activity.
- b) The cost code is not required to post the project resource transaction.
- c) Resource transactions for an inventoried item affects the Project Control Ledger Distribution Journal.
- d) Results of the posted resource transactions are viewed on the Project Resources form, % Complete tab.



Actual project labor costs are not updated until the labor transactions are posted.

- a) True
- b) False



What must be specified when performing a project resource transaction?

- a) Cost code
- b) Project resource
- c) Project task
- d) Source



Which of the following must be entered when creating a voucher adjustment distribution record? Select all that apply.

- a) Cost code
- b) Project number
- c) Project resource
- d) Project task number
- e) Voucher number



Where can you view details of project shipping transactions?

- a) Project Resources form, General tab
- b) Project Resources form, Shipping tab
- c) Project Resource Transactions form
- d) Project Resource Shipping form



What are the three WIP Relief Method options? Select all that apply.

- a) Calculated
- b) Fixed Price
- c) Manual
- d) Revenue Recognition
- e) Shipment



Where can the WIP Relief Method be altered for individual project tasks?

- a) Projects form, WIP Relief tab
- b) Project Tasks form, WIP tab
- c) Project Tasks form – WIP Relieved
- d) Project WIP Adjustments



Which of the following statements apply to creating a revenue milestone? Select all that apply.

- a) If the milestone has not been posted, the Calculate Cost/Revenue button displays above the Planned Cost fields.
- b) Use the Revenue Recognition form to set up revenue milestones.
- c) The WIP Relief Method must be set to Revenue Recognition on the Projects form, WIP Relief tab in order to create a revenue milestone.
- d) There are planned costs, actual costs, and revenue for each revenue milestone.



What are the two steps to print/post a revenue milestone?

- a) Nominate and post the milestone
- b) Nominate and print the milestone
- c) Nominate and process the milestone
- d) Print and post the milestone



Where can you change the invoice method for a project?

- a) Projects form, Cost tab
- b) Projects form, General tab
- c) Projects form, Invoicing tab
- d) Project Parameters form



Appendices

The following are included in this section:

- Appendix A: User accounts
- Appendix B: Project reports and utilities
- Appendix C: Accounting references
- Appendix D: Project change order tracking
- Appendix E: Updates made to this version of the workbook

Appendix A: Learner user accounts

Your instructor will assign you a student user ID from the table listed below to use for class exercises.

Note: If you are taking this course as self-directed learning, refer to the Training Desktop Login Instructions on the Lab On Demand page.

Training Environment Entry Point (VM)	ID	User	Password
CSI SL10.18 Training Desktop	ALL	Infor	Infor123
Application	User ID	User name	Password
Train Click Once Client or Train Web Client	train01	train01@gdeinfor2.com	train01
Train Click Once Client or Train Web Client	train02	train02@gdeinfor2.com	train02
Train Click Once Client or Train Web Client	train03	train03@gdeinfor2.com	train03
Train Click Once Client or Train Web Client	train04	train04@gdeinfor2.com	train04
Train Click Once Client or Train Web Client	train05	train05@gdeinfor2.com	train05
Train Click Once Client or Train Web Client	train06	train06@gdeinfor2.com	train06
Train Click Once Client or Train Web Client	train07	train07@gdeinfor2.com	train07
Train Click Once Client or Train Web Client	train08	train08@gdeinfor2.com	train08
Train Click Once Client or Train Web Client	train09	train09@gdeinfor2.com	train09
Train Click Once Client or Train Web Client	train10	train10@gdeinfor2.com	train10
Train Click Once Client or Train Web Client	train11	train11@gdeinfor2.com	train11
Train Click Once Client or Train Web Client	train12	train12@gdeinfor2.com	train12
Train Click Once Client or Train Web Client	train13	train13@gdeinfor2.com	train13
Train Click Once Client or Train Web Client	train14	train14@gdeinfor2.com	train14
Train Click Once Client or Train Web Client	train15	train15@gdeinfor2.com	train15
Train Click Once Client or Train Web Client	train16	train16@gdeinfor2.com	train16

Train Click Once Client or Train Web Client	train17	train17@gdeinfor2.com	train17
Train Click Once Client or Train Web Client	train18	train18@gdeinfor2.com	train18
Train Click Once Client or Train Web Client	train19	train19@gdeinfor2.com	train19
Train Click Once Client or Train Web Client	train20	train20@gdeinfor2.com	train20
Train Click Once Client or Train Web Client	train21	train21@gdeinfor2.com	train21
Train Click Once Client or Train Web Client	train22	train22@gdeinfor2.com	train22
Train Click Once Client or Train Web Client	train23	train23@gdeinfor2.com	train23
Train Click Once Client or Train Web Client	train24	train24@gdeinfor2.com	train24
Train Click Once Client or Train Web Client	train25	train25@gdeinfor2.com	train25

Appendix B: Reports and utilities

Project reports

Project contains a variety of reports for user analysis. The following table provides a list and description of the project reports:

Report	Description
Cost to Complete Log Report	This report displays the actual, forecast, and cost to complete values for each cost code on a project by task. The report can be driven by the Cost to Complete Snapshot activity to provide this information as of any date.
Indented WBS Cost Report	This report displays all levels of a WBS with the actual and forecast costs as well as a percent complete.
Invoice Milestone Progress Report	This report displays the actual, forecast, and cost to complete values for each cost code by project task. This report can be driven by the Cost to Complete Snapshot activity as of any date.
Nominated Invoice Milestones Report	This report displays all nominated invoice milestones for a particular project range as well as through a particular accounting period. This report is designed to show what invoice milestones will be posted when running the Invoice Milestone Posting activity.
Nominated Revenue Milestones Report	This report displays all nominated revenue milestones for a particular project range as well as through a particular accounting period. This report is designed to show what revenue milestones will be posted when running the Revenue Milestone Posting activity.
Packing Slip Report	This report generates, modifies, and prints packing slips that are related to shipments performed by the Resource Shipment activity.
Print Project Advanced Ship Notice Report	This report allows printing of any project related advance ship notice.
Project Change Order Detail Report	This report displays a summary of all changes to a project for project types that use project tracking.
Project Margin Analysis Report	This report displays actual costs to date for a project, the projected cost to complete the project, projected revenue, margin, and margin percent.

Report	Description
Project Pre-Ship Packing Slip Report	This report generates, modifies, and prints packing slips based on the quantity of project resources requested for the project rather than printing the report based on actual shipment quantity.
Project Purchase Order Commitments Report	This report lists all purchase orders that are cross-referenced to project resources and all purchase orders created for job orders that are cross-referenced to project resources. Purchase orders print by project task and display quantity ordered, quantity received, and cost for each purchase order.
Project Transactions Report	This report displays posted transactions for projects and project tasks by a selected transaction date range or cost code criteria.
Project Variances Report	This report displays estimated, planned, and actual costs by project and project task. The user can choose which variances will display: planned cost vs. actual cost, or estimated cost vs. actual cost.
Project WIP Valuation Report	This report displays the status of project WIP values for each project as well as a product code range. It also displays the actual WIP in the project, the WIP relieved from the project, and the net WIP balance for each project.
Project WIP Valuation by Account Report	This report summarizes all WIP values by account for a particular range of projects.
Reprint Project Packing Slip Report	This report allows reprinting of a project related packing slip.
Revenue Milestone Progress Report	This report displays all revenue milestones for a particular project range and their related statuses.
Unposted Project Labor Transactions Report	This report displays all of the unposted labor transactions by employee date and transaction date. This can be used as an edit list to verify that labor entries were correctly entered.
WBS Percent Complete Report	This report displays the actual and forecast costs for each component of the WBS with the percent complete.

Project utilities

Project contains a variety of utilities for database management. The following table describes two commonly used project utilities:

Utility	Description
Roll Up Project Cost	This utility allows the roll up of all costs and revenues for a project based on several options.
Project Roll From Forecast	This utility allows the user to set budget cost code field values from the forecast field values.

Appendix C: Accounting references

Project Control Ledger Distribution Journal references

Journal reference	Description
PRJ MTL – Project Resource Transaction	This reference occurs when a project resource transaction is performed.
PRJ ADJ – Project Adjustment Transaction	This reference occurs when an adjustment transaction is posted.
PRJ LBR – Project Labor Transaction	This reference occurs when a project labor transaction is posted
JCLS PRJ – Project Job Order Other WIP Variances	This reference occurs when a job is cross-referenced to a project and is closed with WIP variances.
JFIN PRJ – Project Job Order Finish	This reference occurs when a job is cross-referenced to a project and an item has been partially completed on a job and moved to inventory without completing the job.
JOL PRJ – Project Job Order Labor	This reference occurs when a job is cross-referenced to a project and a job labor transaction has been posted to a job.
JOM PRJ – Project Job Order Material	This reference occurs when a job is cross-referenced to a project and a job material transaction has been posted to a job.
JMUV PRJ – Project Job Order Material Usage Variances	This reference occurs when a job is cross-referenced to a project; the application is set for standard costing in system parameters; the material issued to the job is greater than or less than the standard quantity required; and the job in question is complete.
JFMO PRJ – Project Job Order Fixed Material Overhead Variances	This reference occurs when a job is cross-referenced to a project; the job shop floor parameter labeled “Ovhd Basis” contains at least material; the Fixed Matl Ovhd Rate in the Product code form is set to a value greater than zero for those items found on a job BOM; and the job BOM items have been issued to the job.

Appendix D: Project change order tracking

Project Change Orders

Changes to a project can be tracked using the **Project Change Orders** form.

- Select the **Track Changes** check box on the **Projects** form when you create a new project.
- Select “Tracking” on the **Project Types** form.

Access the **Project Change Orders** form by selecting **Master Explorer > Roles > Project Management > Project Change Orders**.

The following table describes the fields, drop-down lists, and grid on the **Project Change Orders** form:

Field / grid	Description
Project	The Project drop-down list displays the projects with the Track Changes check box selected on the Projects form.
Change	The Change field displays the Change Number.
Change Status	The Change Status drop-down list displays the following options: • Opened • Finalized • Printed
Change Date	The Change Date field displays the last change date.
User Name	The User Name field displays the User Name for the last change.
Project Resource Change Detail	The Project Resource Change Detail displays change detail in a grid view.

Project Change Order Detail report

The **Project Change Order Detail Report** displays a summary report of all changes to a project.

Access the **Project Change Order Detail Report** by selecting **Master Explorer > Roles > Project Management > Reports > Project Change Order Detail Report**.

Appendix E: Updates made to this version of the workbook

Summary of updates

We updated the versioning of the document and made the following content changes.

Updates to the training experience

- Updated the section on User interface to add web client and smart client login information

Updates for new features

- Updated the section on User interface to add web client and smart client login information
- Lesson 3 – added transfer order source information for project resources
- Lesson 13 - New translate icon for multi-currency users to convert project costs and revenue to domestic currency